

Supplementary information

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Supplementary Sequences 1. Amino acid sequences for cytidine deaminase and adenosine deaminase of evolved base editor in this study.

Supplementary Table 1 Materials used.

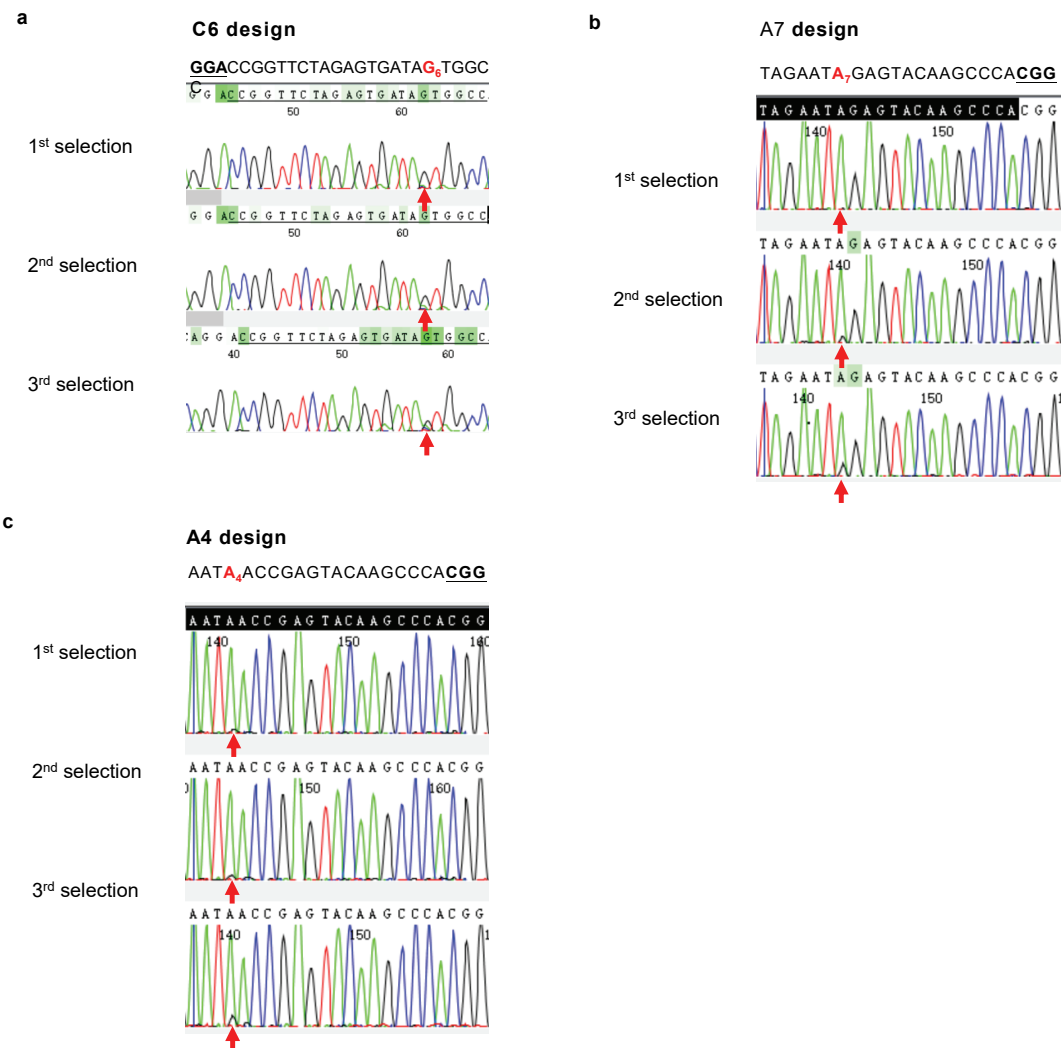
Supplementary Table 2. Comparison of ABE variants.

Supplementary Table 3. Comparison our CDEM to PACE and directional screening in human cells.

Supplementary protocol: Evolution of BE3 as an example

Supplementary Discussion

References

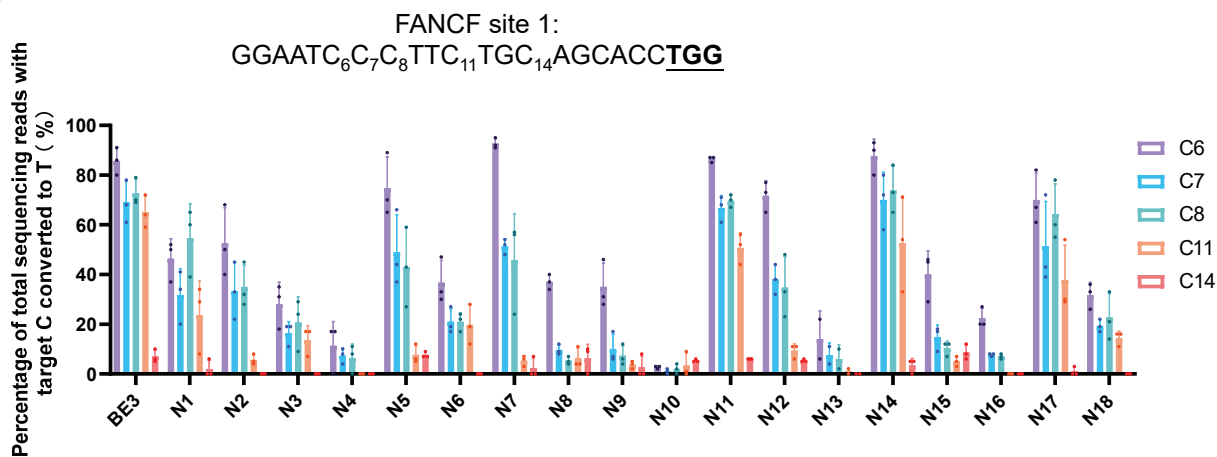


Supplementary Fig 1. Confirmation of selection plasmid editing during evolution. (a) Editing efficiency of selection plasmid in C6 design by BE3 at different selection stages. (b, c) Editing efficiency of selection plasmid in A7 or A4 design by ABE8e or ABEmax at different selection stages.

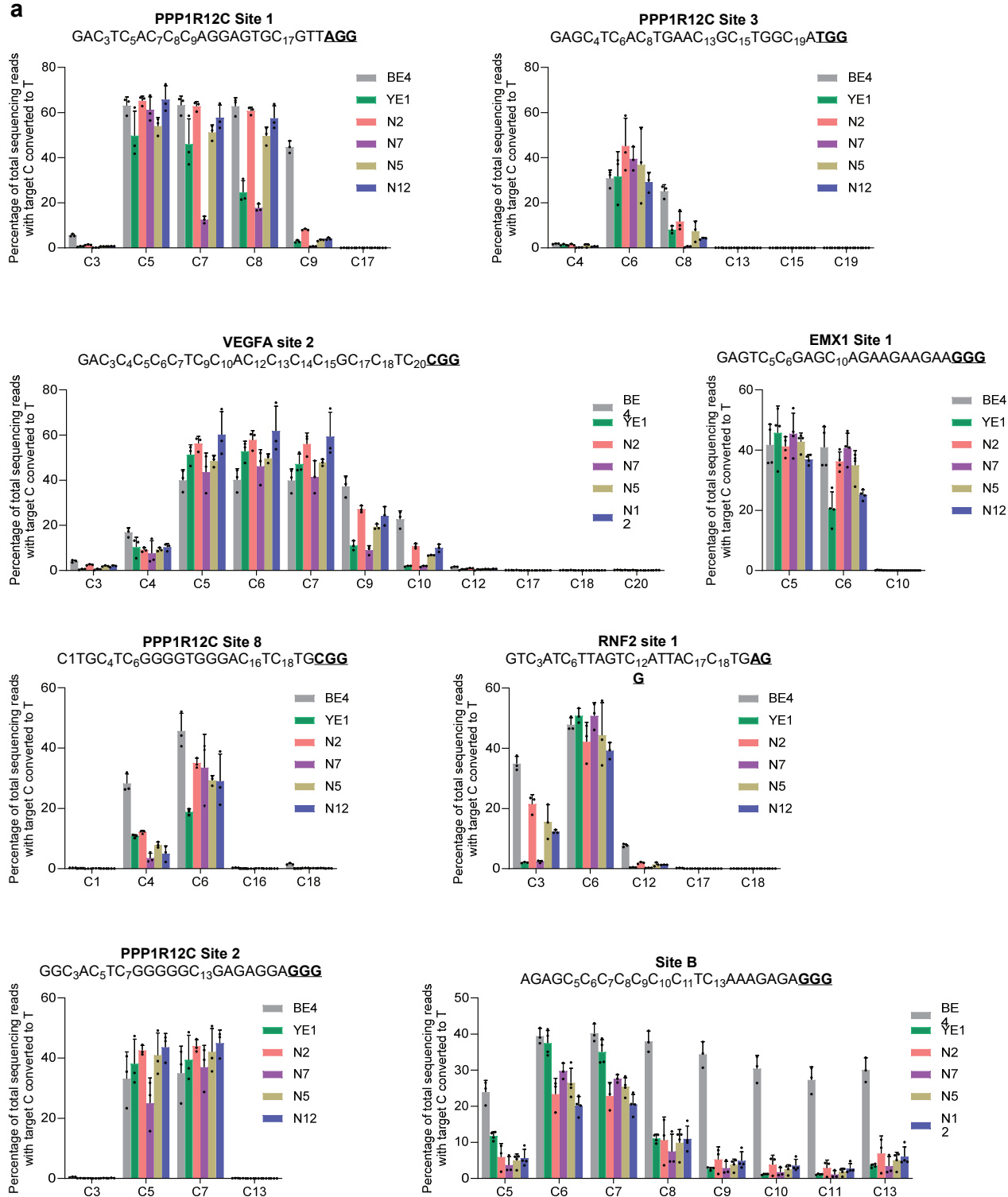
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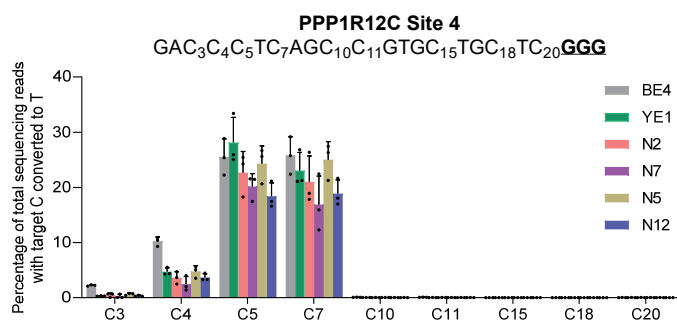
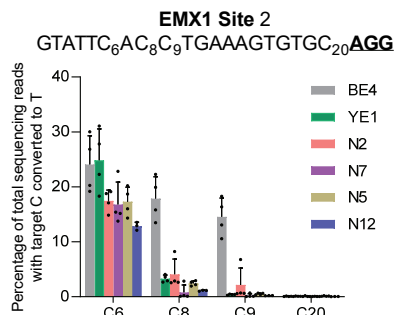
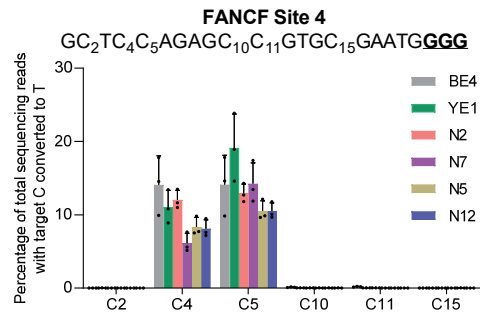
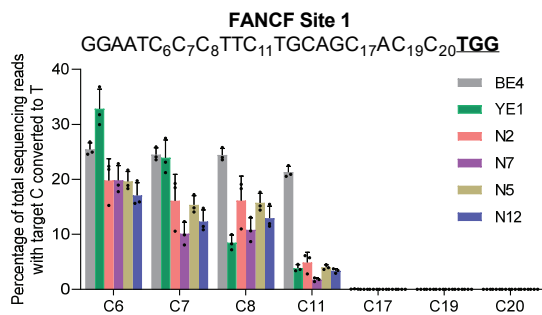
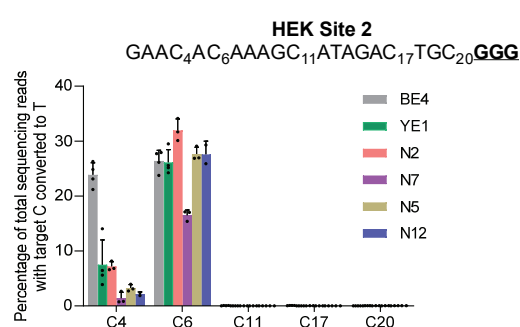
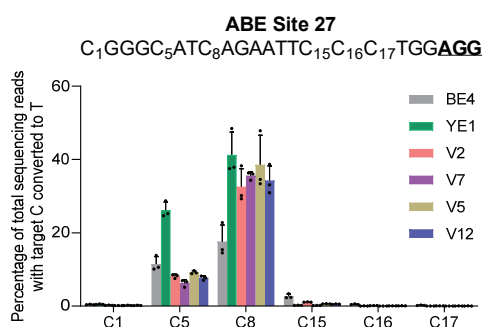
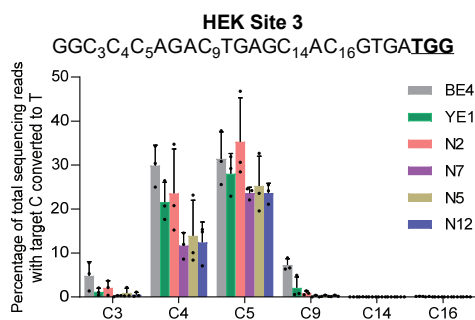
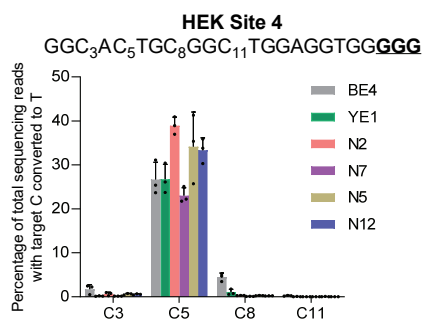
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	BE3	Clone #	T	I	E	L	Q	F	Y	A	Y	L	S	V	T	E	V	L	P	L	R	L	Nonsense mutation
C6-5	N1	1													A	G							P168P(CCT/C)
C6-5	N2	2													A	G		P					
C6-5	N3	1							C						A	G							
C6-5	N4	1		G		G									A	G							
C6-5	N5	4		G		G									A	G							K17K(AAA/G)
C6-5	N6	1		I		V																	
C6-5	N7	7									H												T80T(ACA/G),L104L(C/TTG)
C6-5	N8	8												P									
C6-5	N9	1												P									I67I(ATC/A)
C6-5	N10	1						R						P					P				L134L(T/CTG)
C6-5	N11	1							S												G		
C6-5	N12	7													M						G		S137S(TCA/G)
C6-5	N13	1					P																V25V(GTA/G)
C6-5	N14	1								S													L134L(T/CTG)
C6-5	N15	1										S											
C6-5	N16	3															L						L134L(T/CTG),V139V(GTG/T)
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b

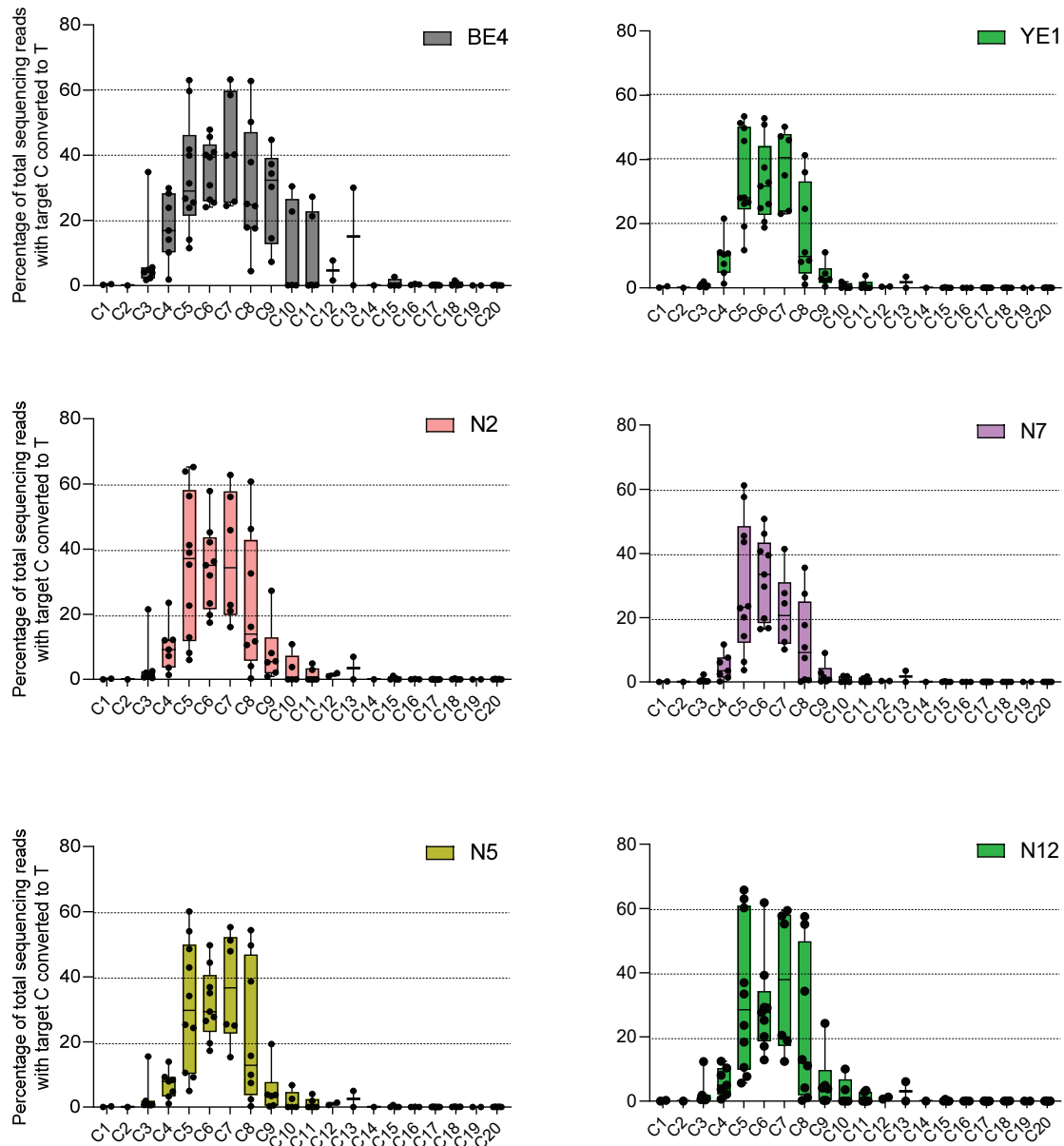


Supplementary Fig. 2 | Genotypes of evolved cytidine deaminase and editing efficiencies at genomic site FANCF1 (a) Genotypes of evolved cytidine deaminase described in this work. (b) Base editing efficiencies of additional evolved BE3 variants at genomic site FANCF 1. The predicted editing bar plot based on Sanger sequencing chromatograms by EditR. Bars represent mean values, and error bars represent the s.d.

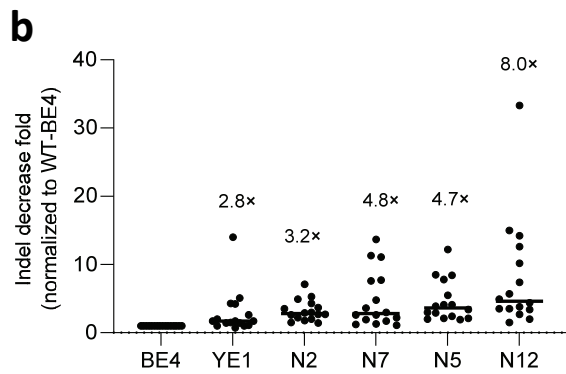
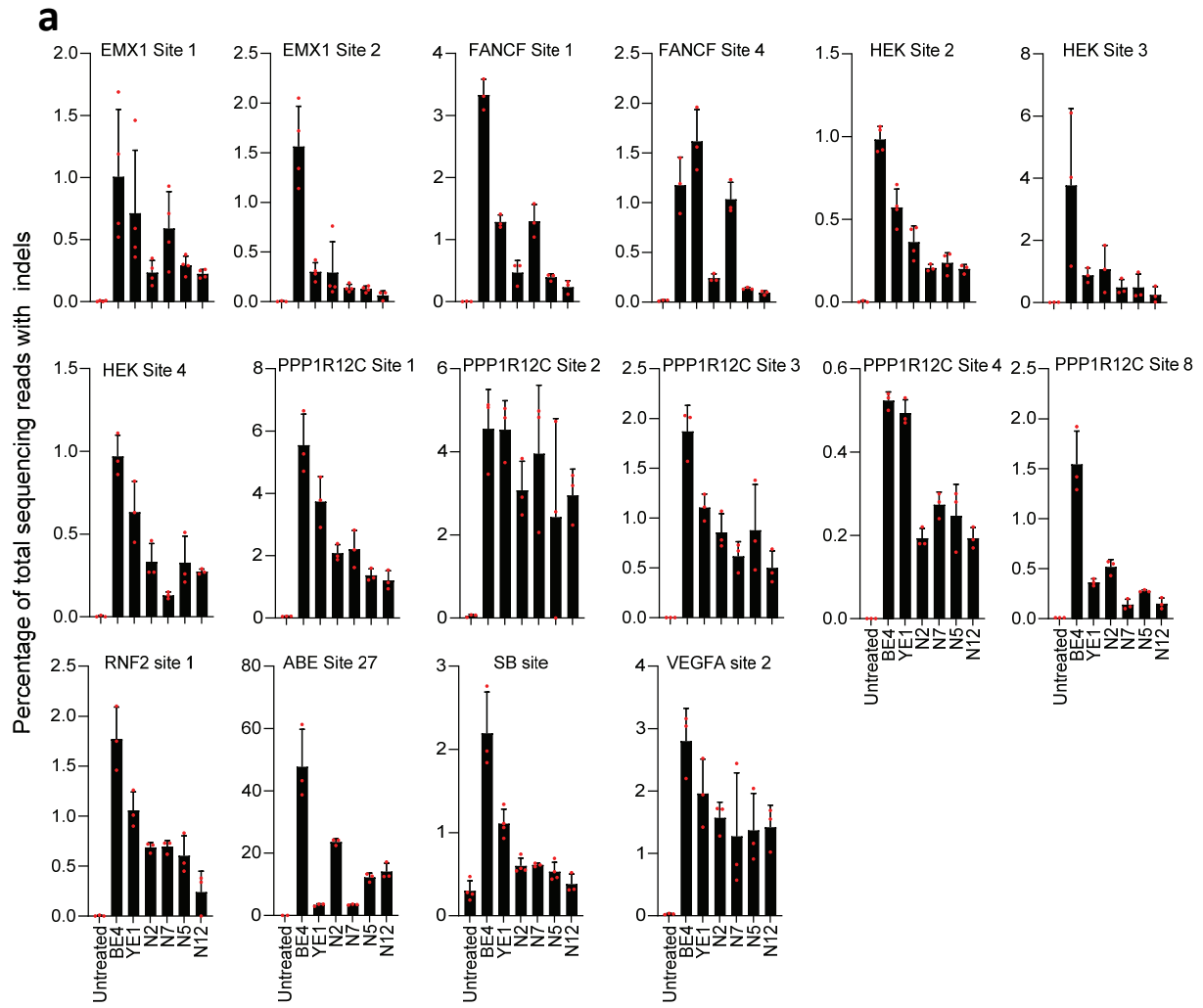
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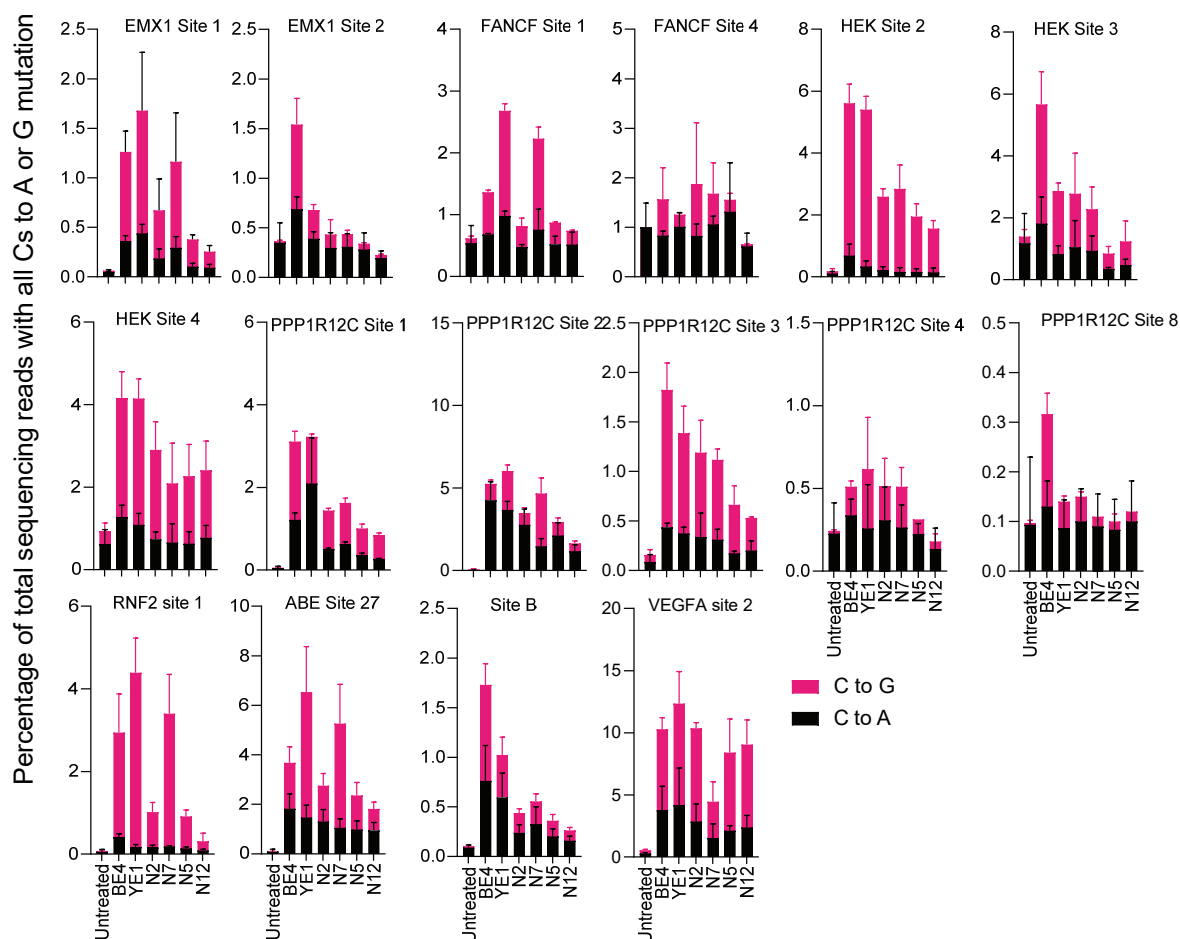
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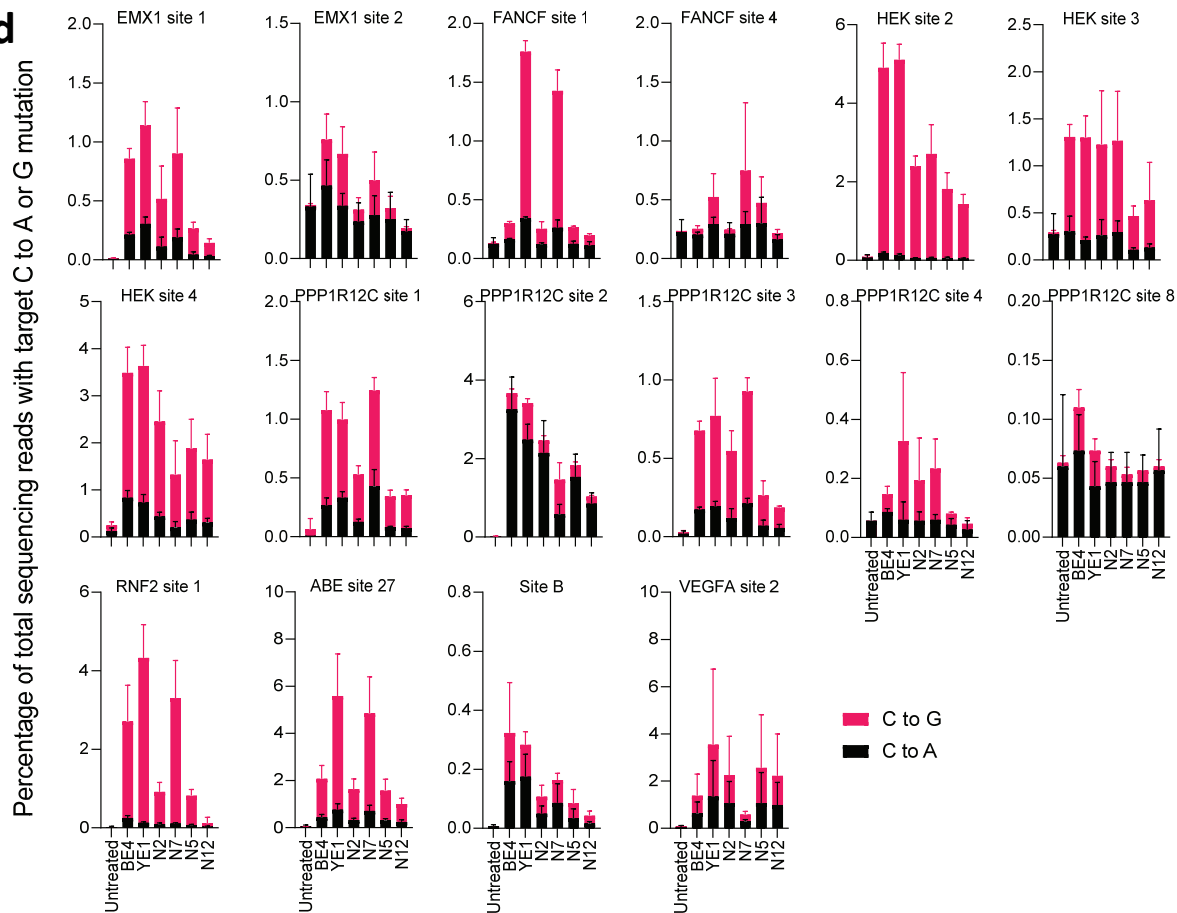
Supplementary Fig. 3. Characterization of more evolved cytidine deaminase variants and their editing windows. (a) Evolved CBE variants exhibit comparable editing efficiency and narrower editing windows compared with BE4 or YE1. Bars represent mean values, and error bars represent the s.d. of three independent biological replicates. (b) Frequencies of C-to-T by BE4, YE1 and evolved variants across the protospacer from the edited sites in a (where PAM is at positions 21–23). Single dot represents individual replicate (n = 3 independent replicates per site). Boxes span the interquartile range (25th to 75th percentile); horizontal lines indicate the median (50th percentile); and whiskers extend to minima and maxima.



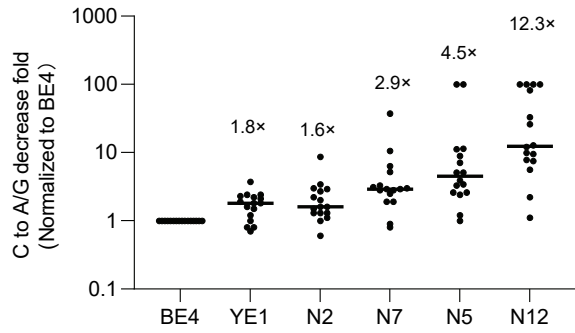
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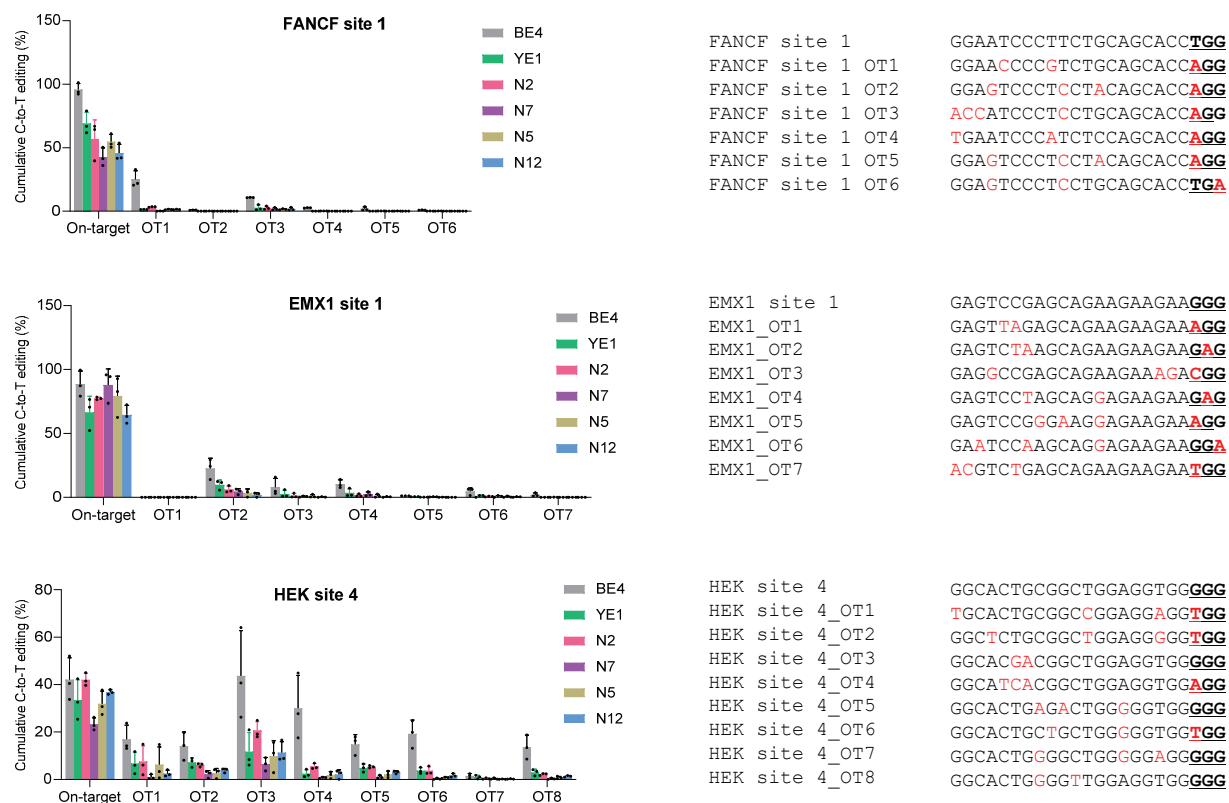


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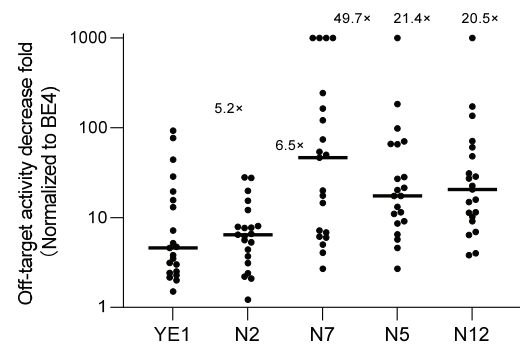


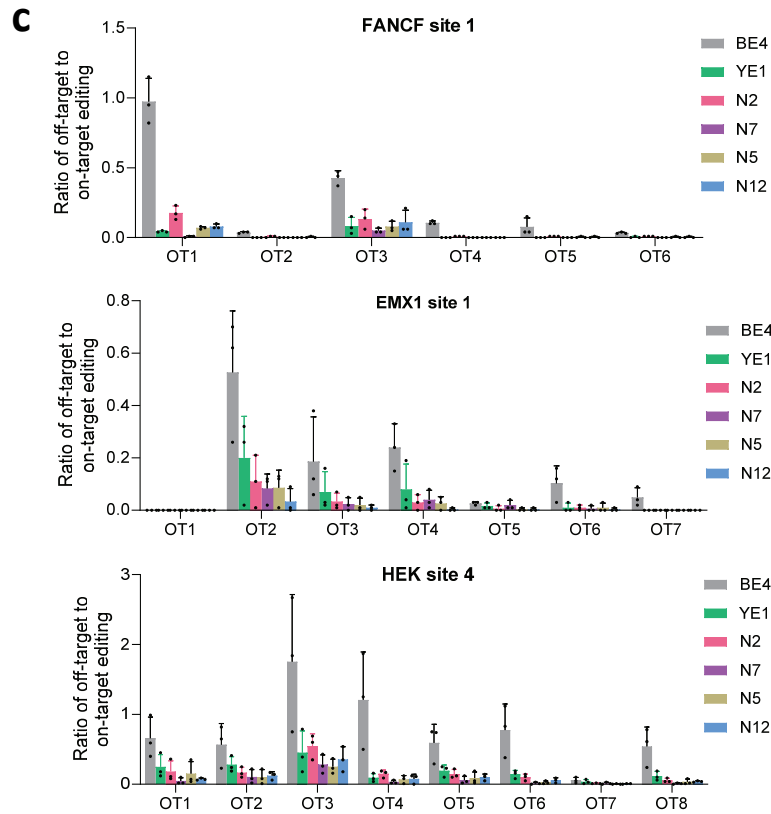
Supplementary Fig. 4 | Evolved CBE variants with higher product purity. (a) HEK293T cells were treated with BE4, YE1 and 4 evolved variants at 16 genomic sites, as described in Materials and Methods. Frequency of indel formation following the treatment was shown. (b) CBE variants YE1, N2, N7, N5 and N12 indel decrease fold relative to BE4 at 16 endogenous target sites in HEK293T cells. Mean indel decrease fold are shown (n = 16 target sites). (c) Targeted all Cs in the spacer converted to A or G of total sequencing reads were shown. (d) Targeted C converted to A or G of total sequencing reads were shown. Targeted C position is the position with highest editing activity of BE4. (e) All Cs in the spacer converted to A or G of evolved variants and YE1 decrease fold relative to BE4 at 16 endogenous target sites in HEK293T cells. The maximal fold changes number were set as 100. Median C converted to A/G decrease fold are shown (n = 16 target sites).

a

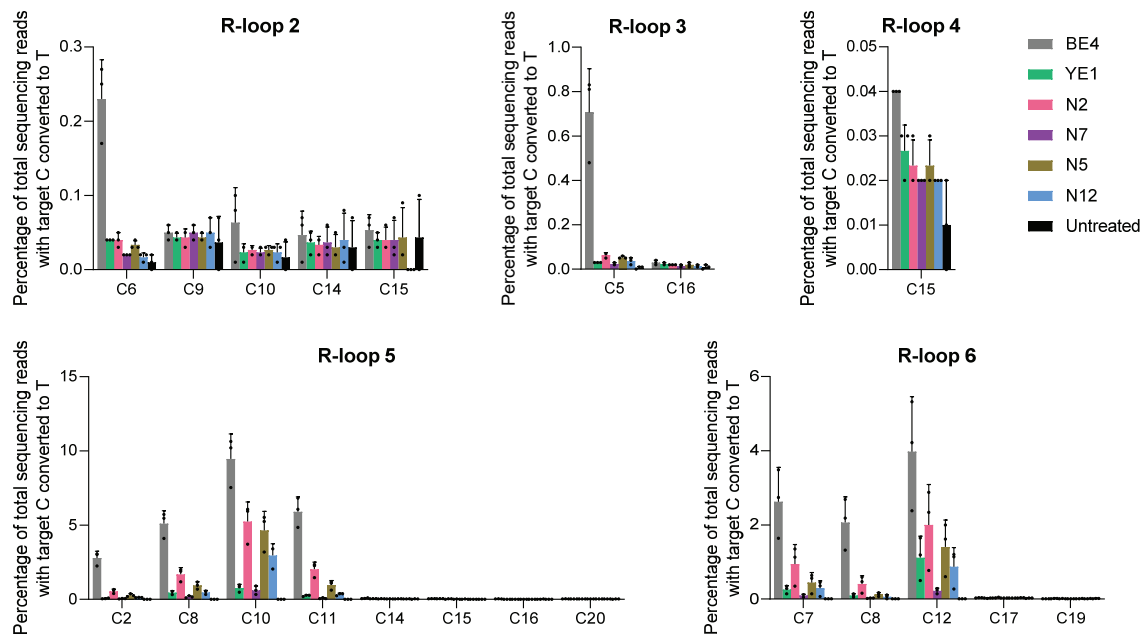


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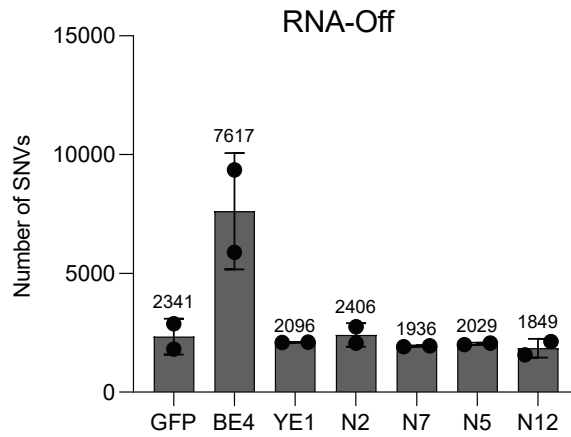


Supplementary Fig. 5 | Evolved variants exhibit lower off-target activity. (a) Off-target editing activities of BE4, YE1 and evolved variants. On- and off-target editing frequencies of three gRNAs targeted to the EMX1 site 1, HEK site 4, FANCF site 1 with BE4, YE1 and evolved variants. Percentage of edits represents the sum of all edited Cs in the editing window (1-15, PAM:21-23) and represent the mean of $n = 3$ biologically independent samples with error bars representing s.d. (b) The off-target activity decrease fold of evolved variants and YE1 relative to BE4 were calculated at all off-target sites at a. Bars represent median. (c) The ratio of off-target (all C) over on-target (single highest C) C-to-T editing efficiencies are shown. Error bars representing s.d.

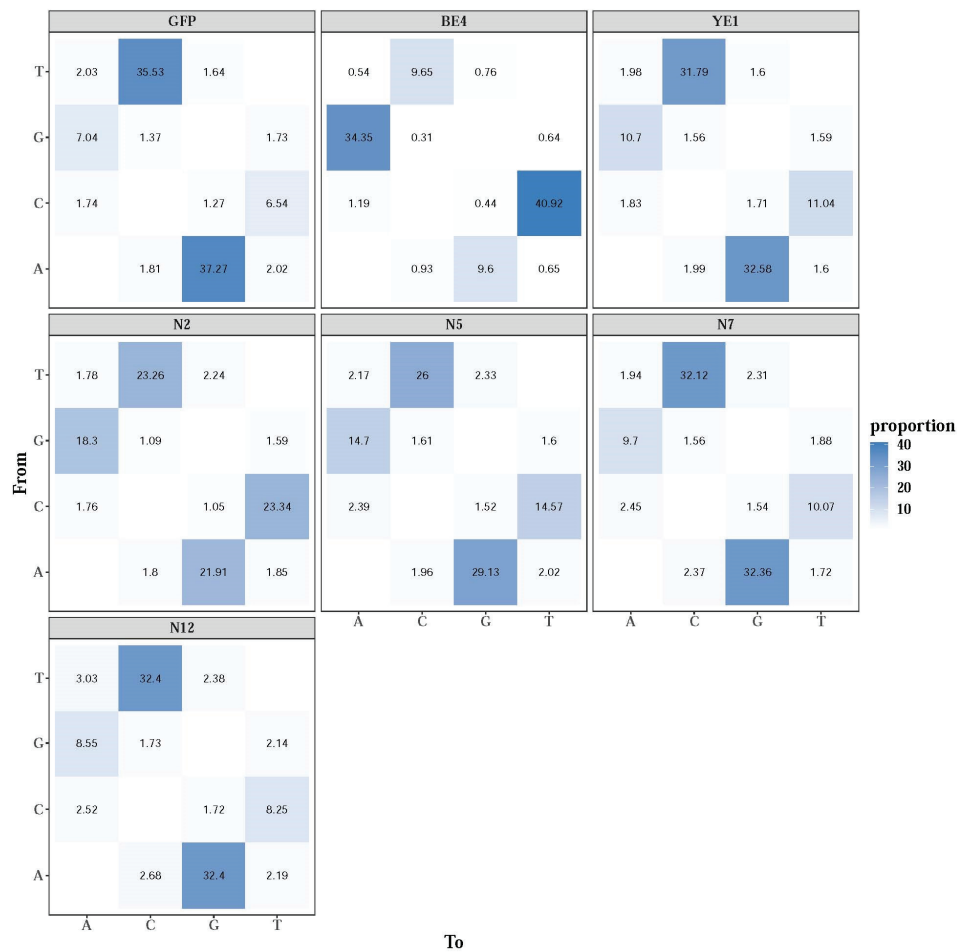


Supplementary Fig. 6 | Cas9-independent deamination by cytidine deaminase in HEK293T cells. Cas9 independent off-target A•T-to-G•C editing frequencies detected by the orthogonal R-loop assay at each R-loop site with dSaCas9 and a SaCas9 sgRNA. Each R-loop was performed by cotransfection of BE4, YE1, and evolved variants, an SpCas9 sgRNA targeting PPP1R12C site 1 with dSaCas9 and a SaCas9 sgRNA targeting R-loops 2–6, respectively.

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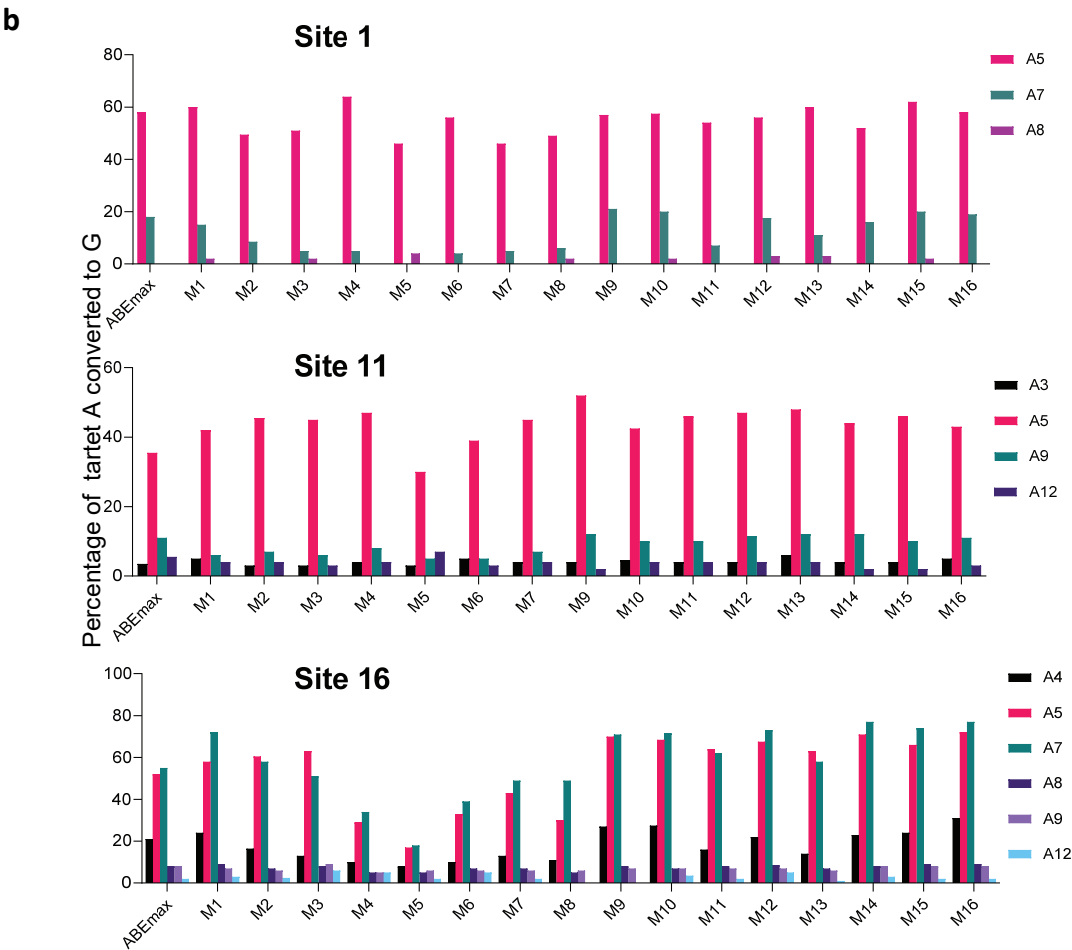
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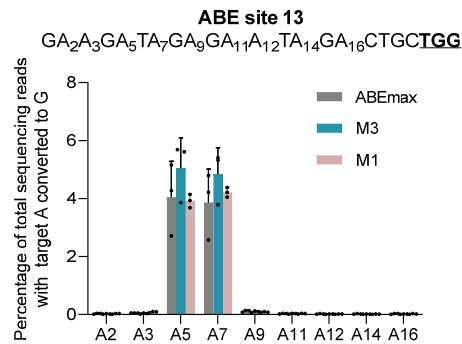
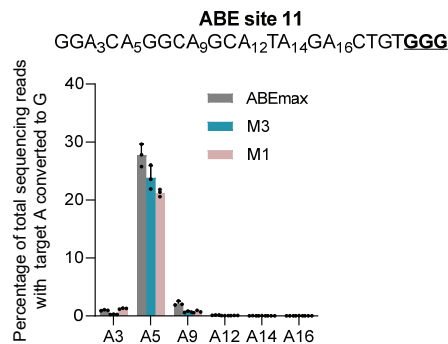
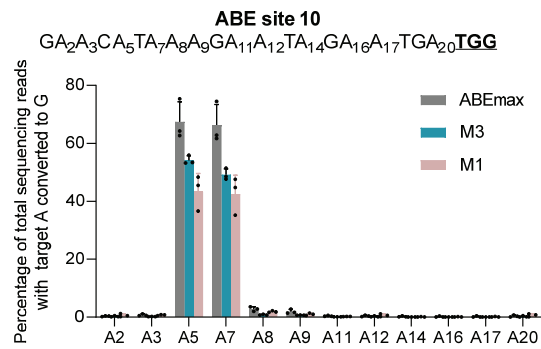
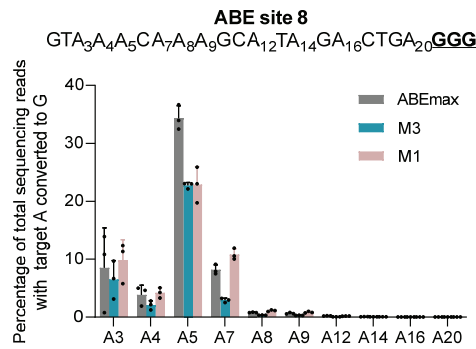
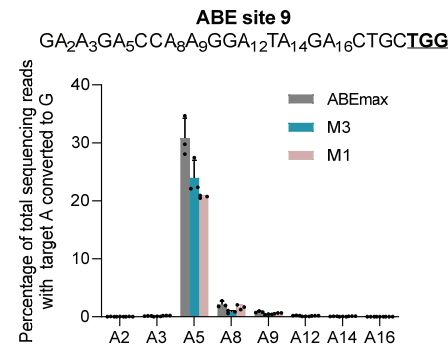
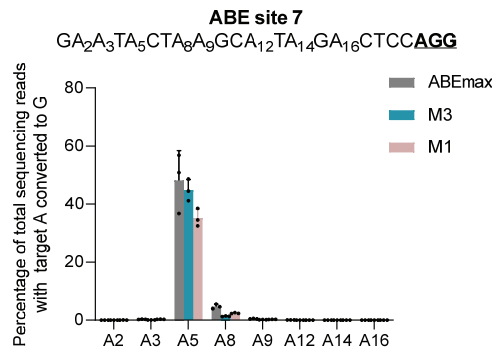
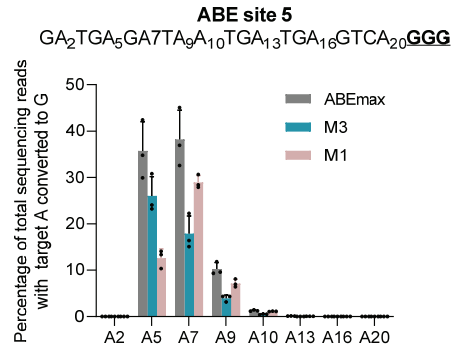
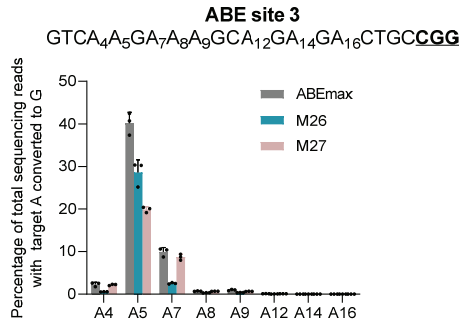
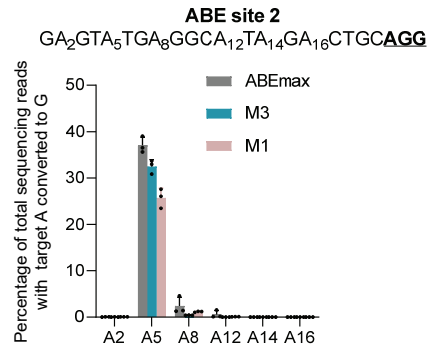
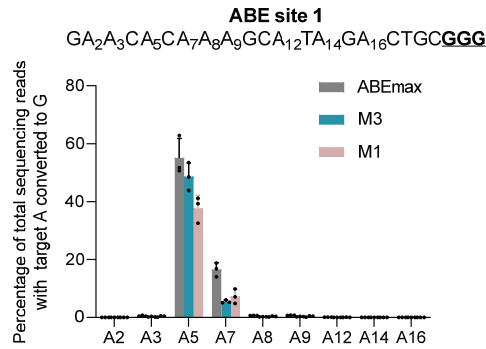
Supplementary Fig. 7| RNA off-target evaluation of evolved variants versus BE4 and YE1. (a) Comparison of the total number of detected RNA off-target SNVs. Bars represent mean values, and error bars represent the s.d. of 2 independent biological replicates. (b) Distribution of mutation types for the groups transfected with BE4, YE1, N2, N5, N7, N12 and GFP. The number indicates the percentage of a certain type of mutation among all mutations.

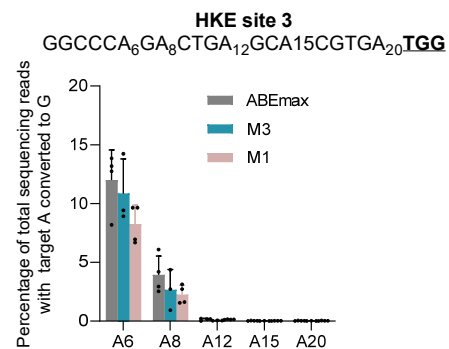
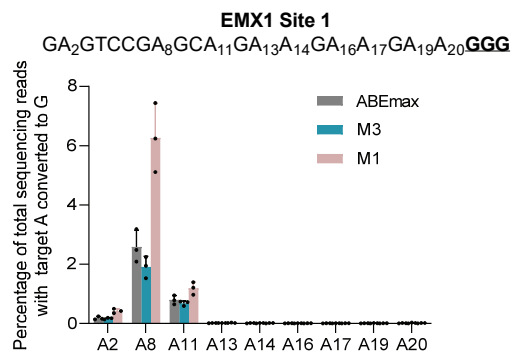
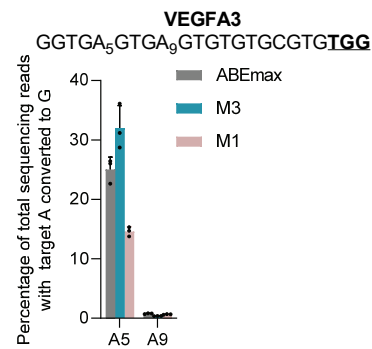
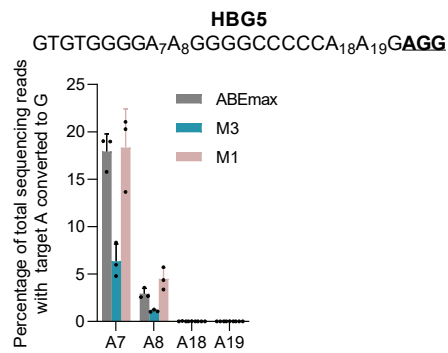
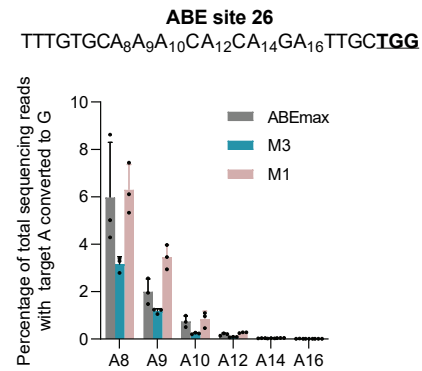
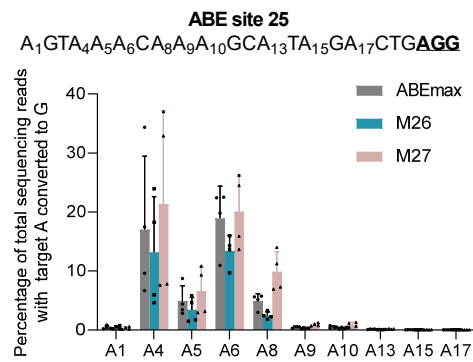
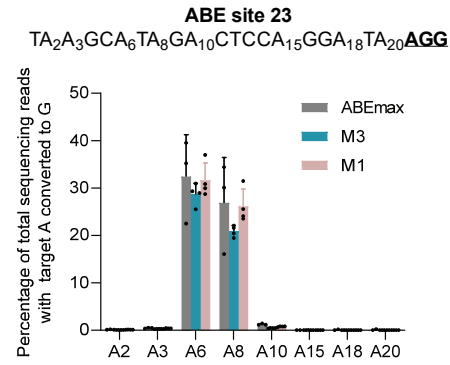
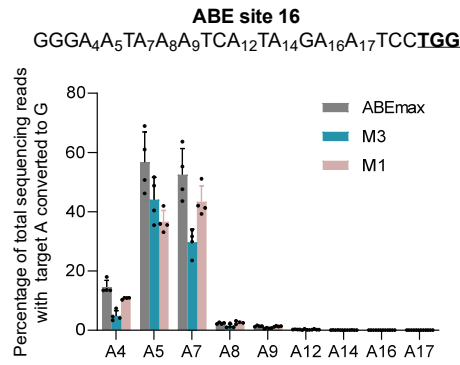
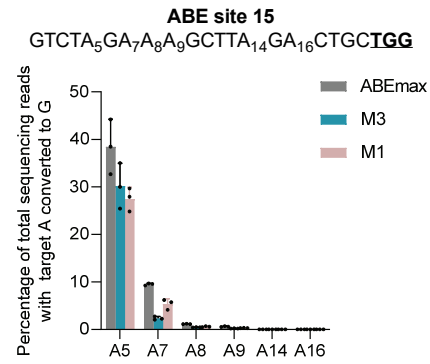
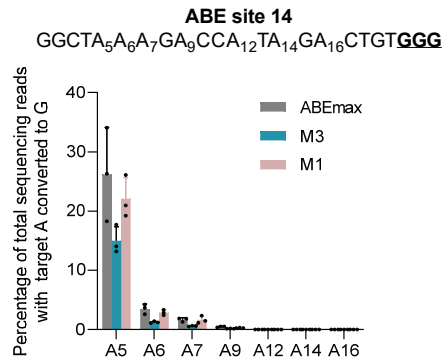
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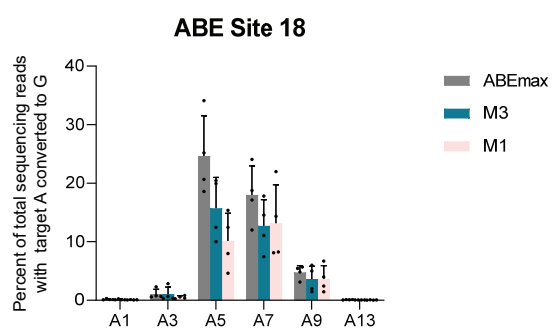
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M1		De137-190												De11-34				Monomer	
M2	2						C												Heterodimer
M3	1	G	C				C											C	Heterodimer
M5	1		C				C	G		S									Heterodimer
M4	1		C							D									Heterodimer
M6	1			G						D									Heterodimer
M7	1									D	V								Heterodimer
M8	1				G	S	C			D		G	E						Heterodimer
M9	1											G	E						Heterodimer
M10	2											G							Heterodimer
M11	1								G							C			Heterodimer
M12	2								G										Heterodimer
M13	1								G						A		G		Heterodimer
M14	1								G					T (ACA/G)			V		Heterodimer
M15	1								G								V		Heterodimer
M16	1																V		Heterodimer



Supplementary Fig. 8| Genotypes of evolved ABEmax and editing efficiencies. (a) Genotypes of evolved ABEmax described in this work. (b) Base editing efficiencies of additional evolved ABEmax variants at genomic sites including Site1, Site11, Site16 in 293T cells. The predicted editing bar plot based on Sanger sequencing chromatograms by EditR.

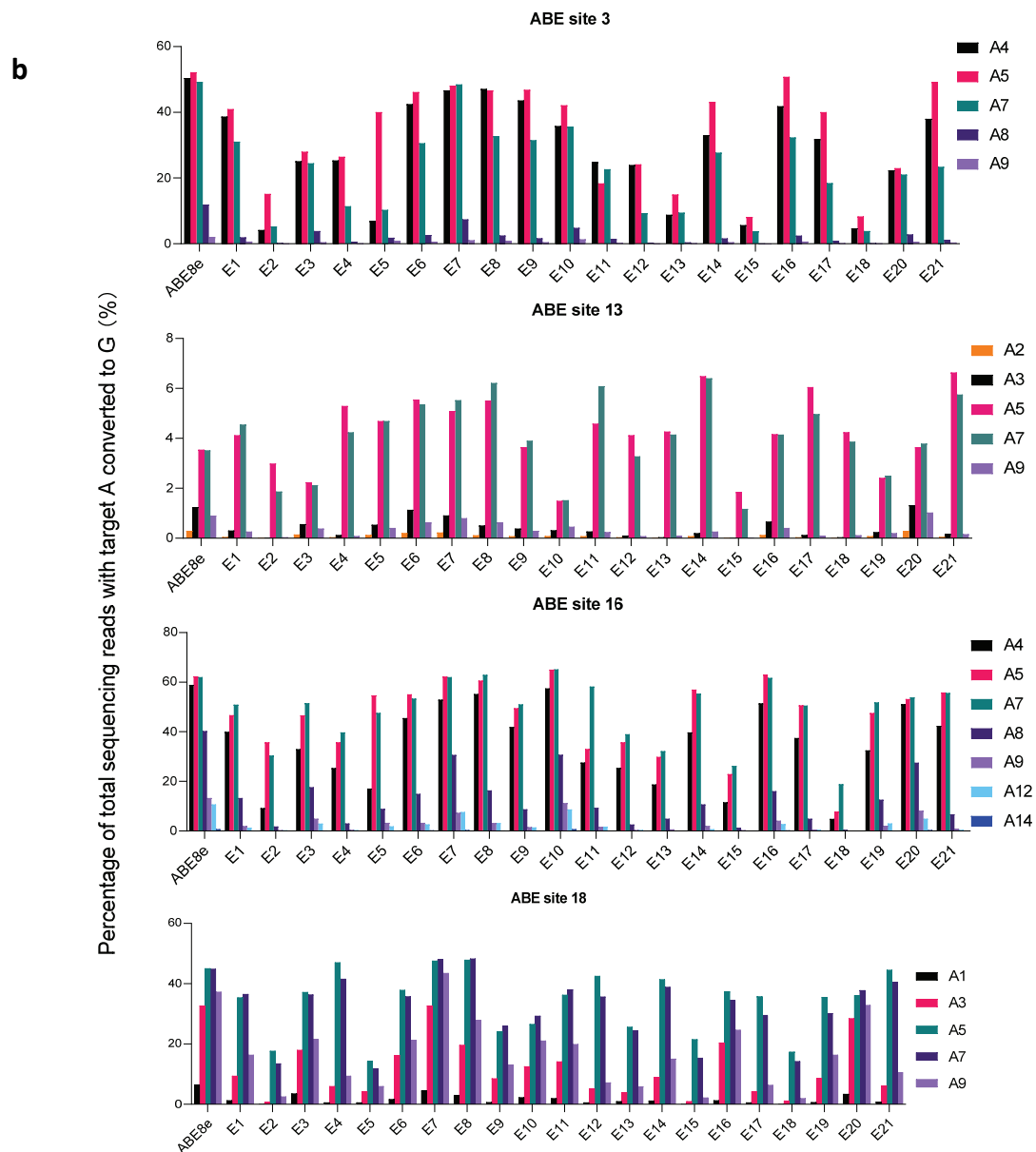






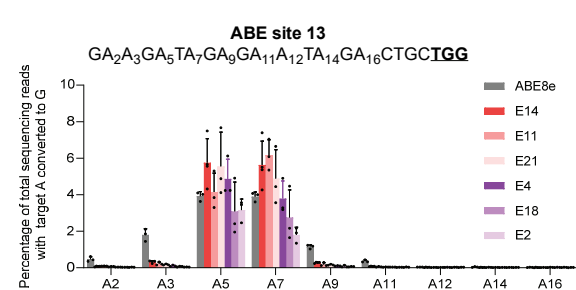
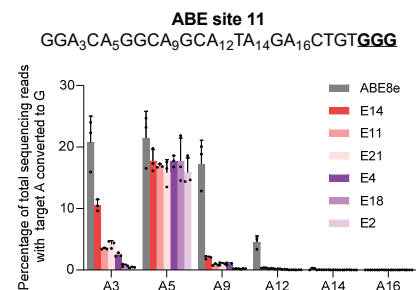
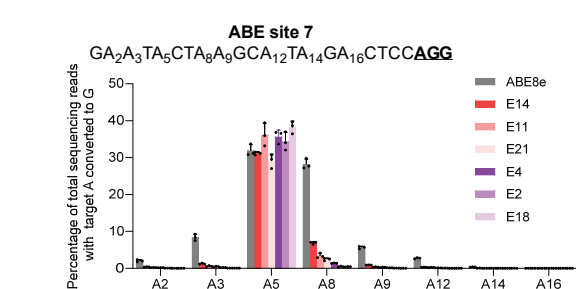
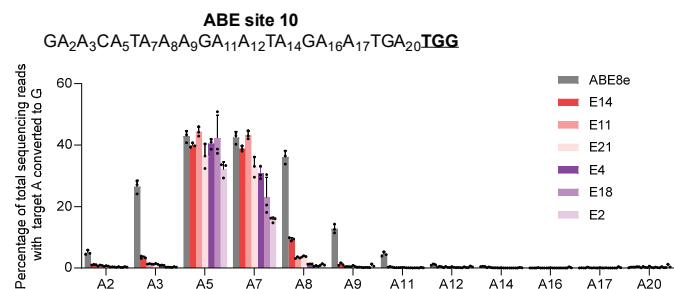
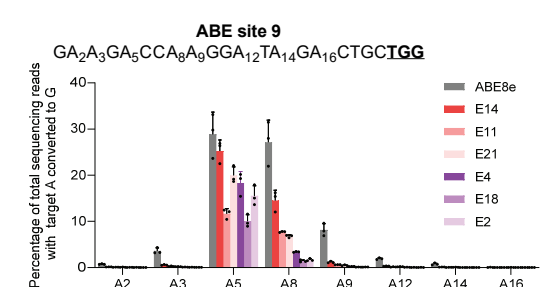
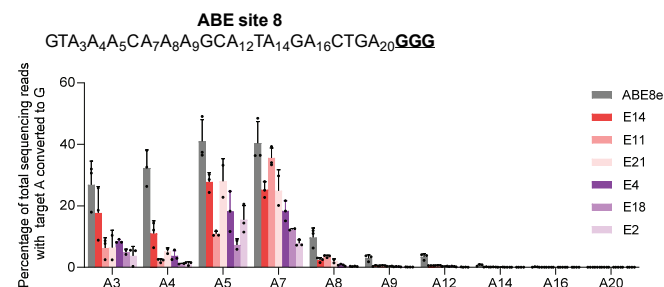
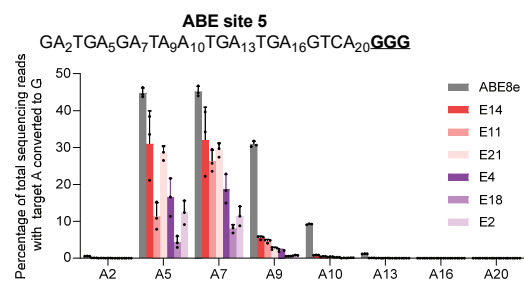
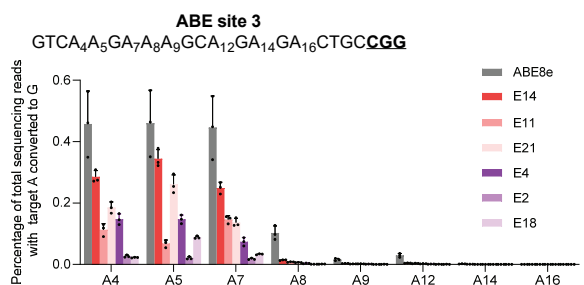
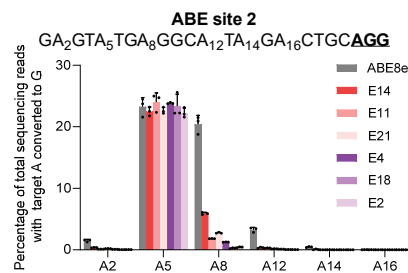
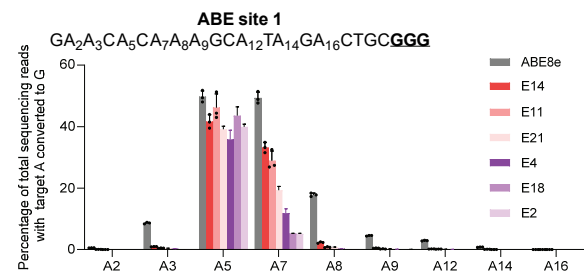
Supplementary Fig. 9| Characterization of more evolved adenosine deaminase variants of ABEmax and their editing windows. Base editing in HEK293T cells by evolved adenosine deaminase on ABEmax architecture.

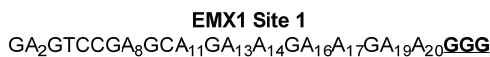
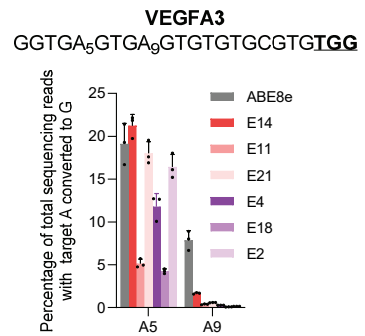
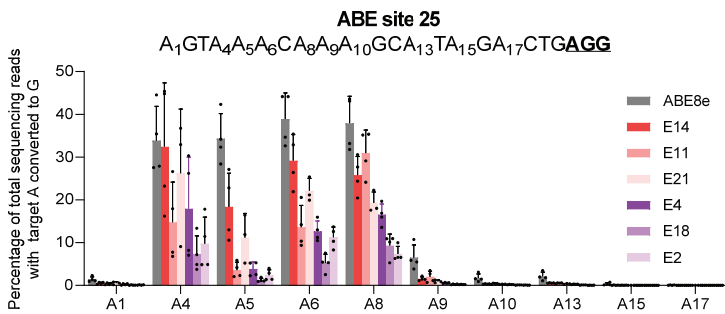
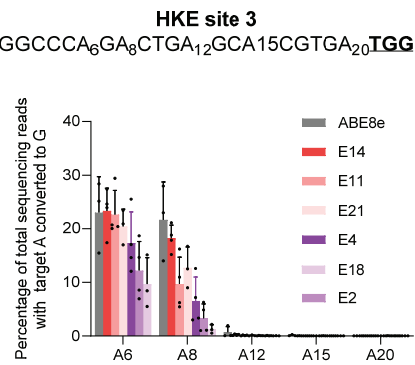
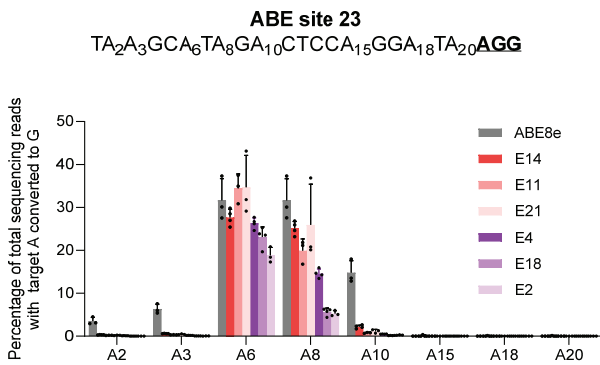
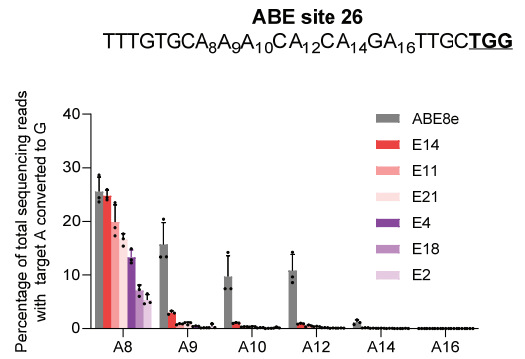
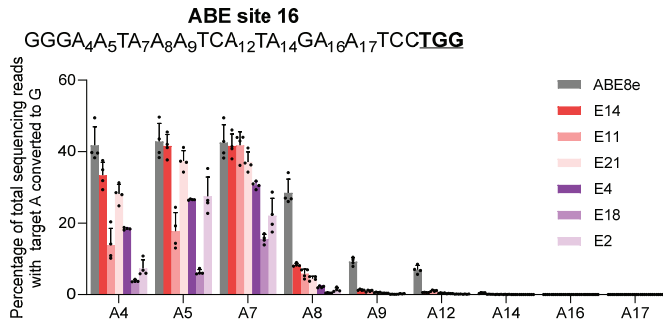
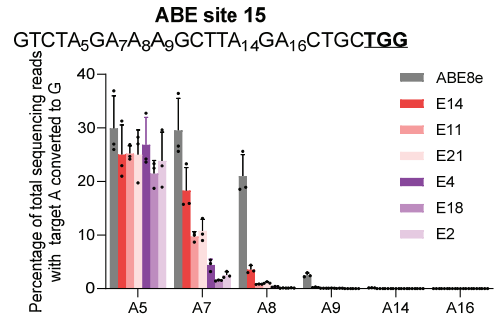
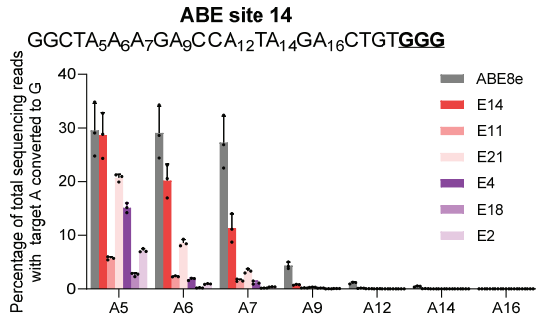
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	A4-5	E2	G				G	G			G		H				V119A		
	A4-4	E3					D			R			C				H51R, I98T, S170C, R63 (AGA/G)		
	A4-5	E4					D					P							
	A4-5	E5					D		P										
	A4-5	E6		G					P								I48 (ATC/T), C140 (TGT/C)		
	A4-5	E7								C				R					
	A4-5	E8								C						G	R152G		
	A4-5	E9											P	C			M60I		
	A7-5	E10	G				D						P	H					
	A7-4	E11											P		R				
	A7-5	E12					G						P						
	A7-5	E13					G	G		R		G					H51R, N121D		
	A7-5	E14					G	G				G				G	G99C		
	A7-5	E15					G	G	P			G					N36S		
	A7-5	E16				G	G	G	P								S6G		
	A7-3	E17	G		G	G			P							D			
	A7-5	E18							P				P			D			
	A7-5	E19							P						R				
	A7-5	E20			G	G								C					
	A7-3	E21		G								G							

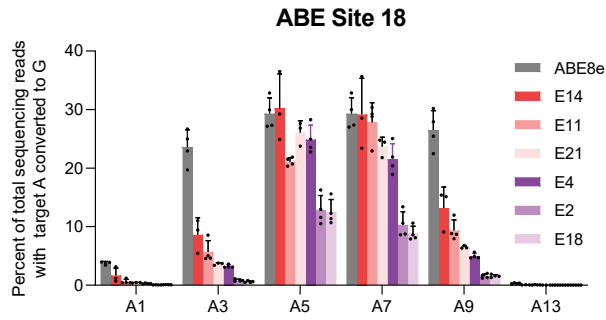


Supplementary Fig. 10 | Genotypes of evolved ABE8e and editing efficiencies. (a) Genotypes of evolved ABE8e described in this work. (b) Base editing efficiencies of additional evolved ABE8e variants at genomic sites including Site3, Site13, Site16, Site18 in 293T cells.

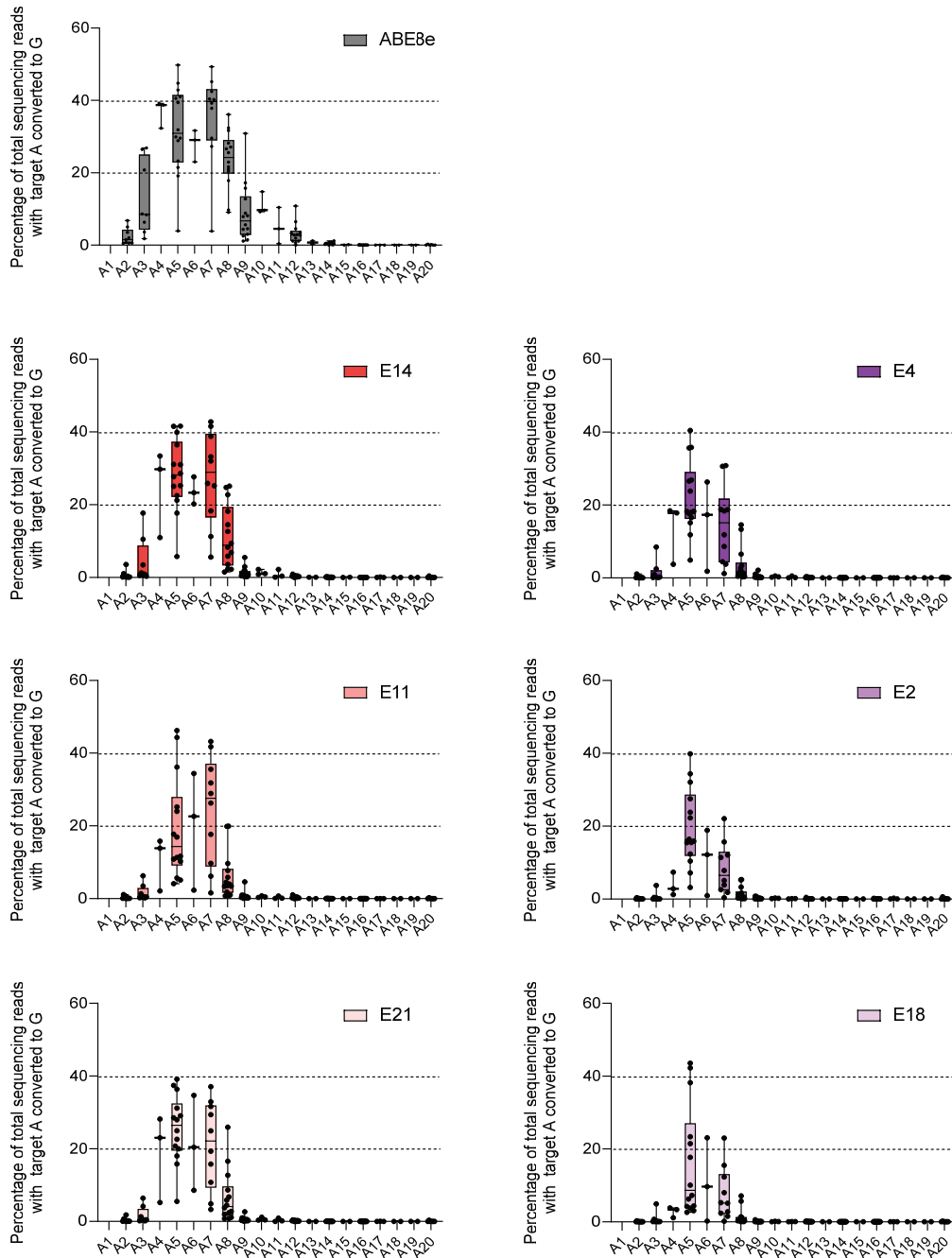
a



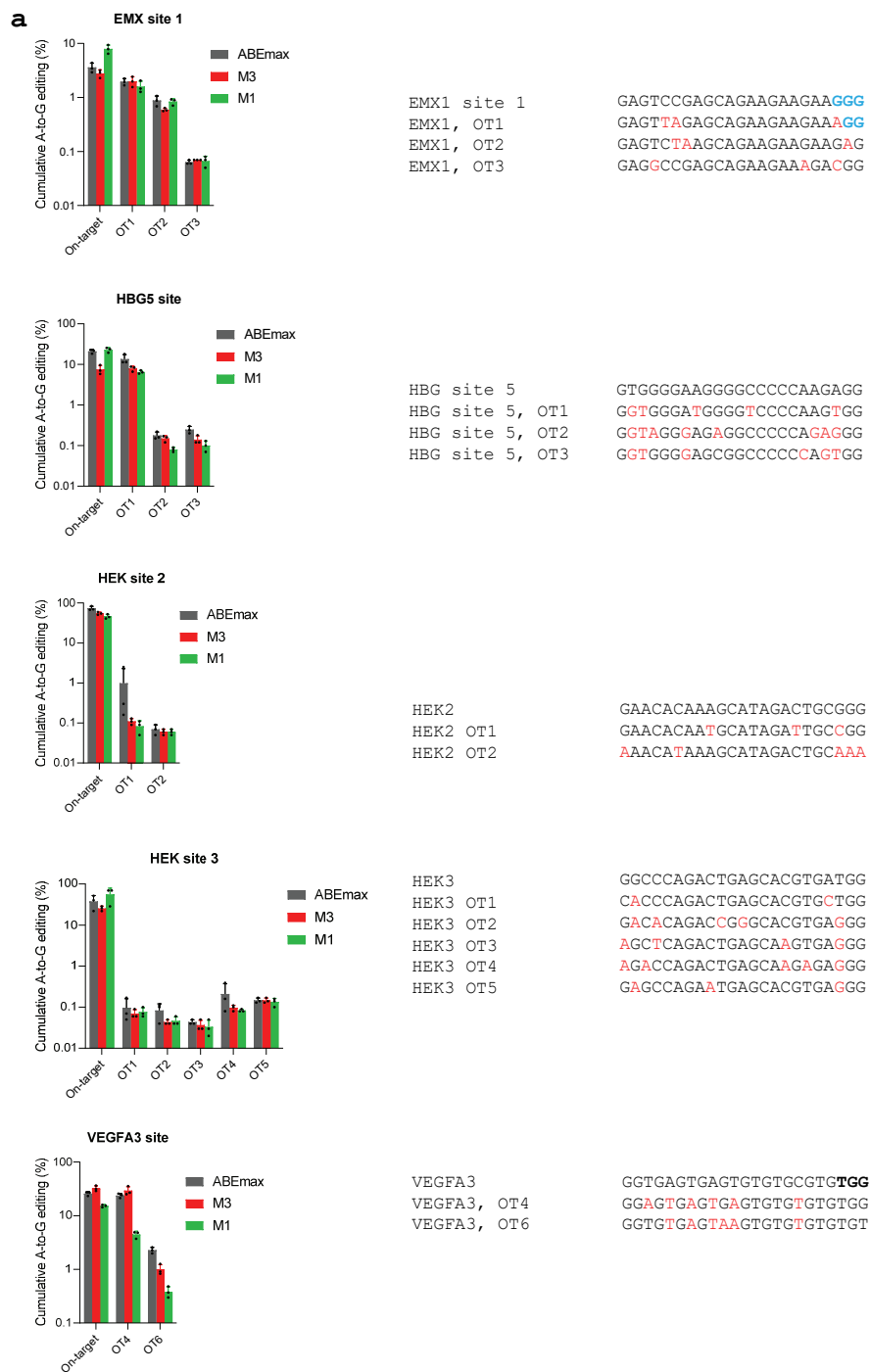




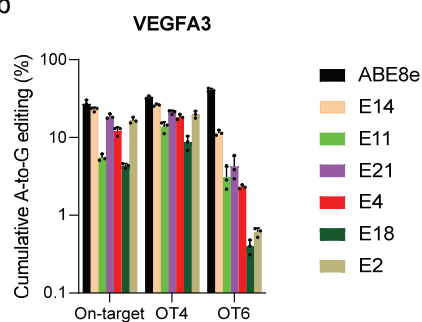
b



Supplementary Fig. 11| Characterization of more evolved adenosine deaminase variants of ABE8e and their editing windows. (a) Base editing in HEK293T cells by evolved adenosine deaminase on ABE8e architecture. (b) Frequencies of A-to-G editing of evolved adenosine deaminase across all protospacer positions



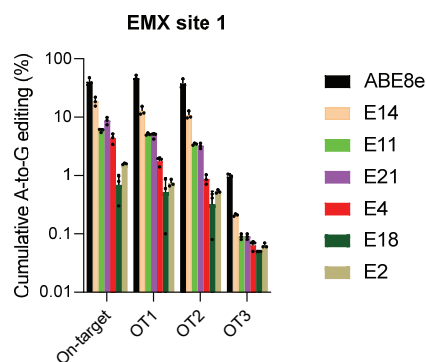
b



VEGFA3

VEGFA3, OT4

VEGFA3, OT6

GGTGAGTGAGTGTGTGCGT**G**TGGGG**A**GT**G**AGT**G**AGTGTGTGTGTGGGGTGT**G**AG**T**AA**G**TGTGT**G**TGTGTGT

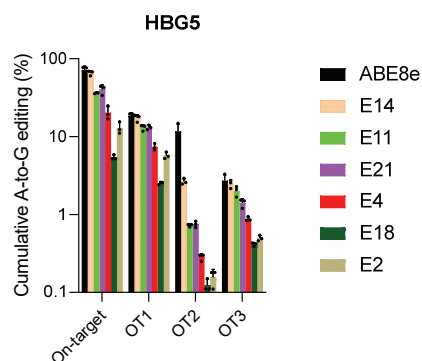
EMX1 site 1

EMX1, OT1

EMX1, OT2

EMX1, OT3

GAGTCCGAGCAGAAGAAGAAGGG

GAGT**T**AGAGCAGAAGAAGAA**A**AGGGAGTC**T**AAGCAGAAGAAGAAG**A**GGAG**G**CCGAGCAGAAGAA**A**G**A**CGG

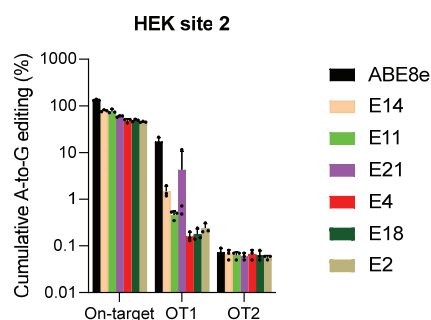
HBG site 5

HBG site 5, OT1

HBG site 5, OT2

HBG site 5, OT3

GTGGGGAAGGGGCCCCCAAGAGG

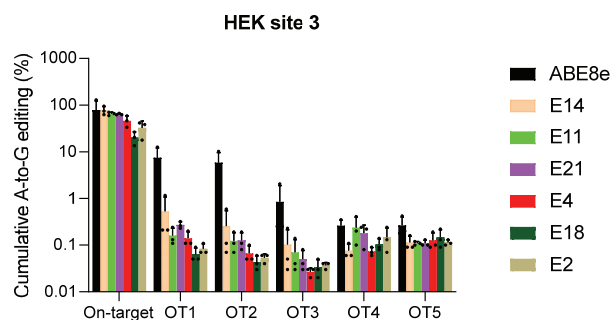
GGTGGG**A**TGGGG**T**CCCCAAG**T**GG**G**GTAGGG**G**AG**A**GGCCCC**C**AGAGGG**G**GTGGG**G**AGCGGCCCC**C**AG**T**GG

HEK2

HEK2 OT1

HEK2 OT2

GAACACAAAGCATAGACTGCGGG

GAACACA**A**TGCATAG**A**TTG**C**CGG**A**AACATAAAGCATAGACTGC**A**AA

HEK3

HEK3 OT1

HEK3 OT2

HEK3 OT3

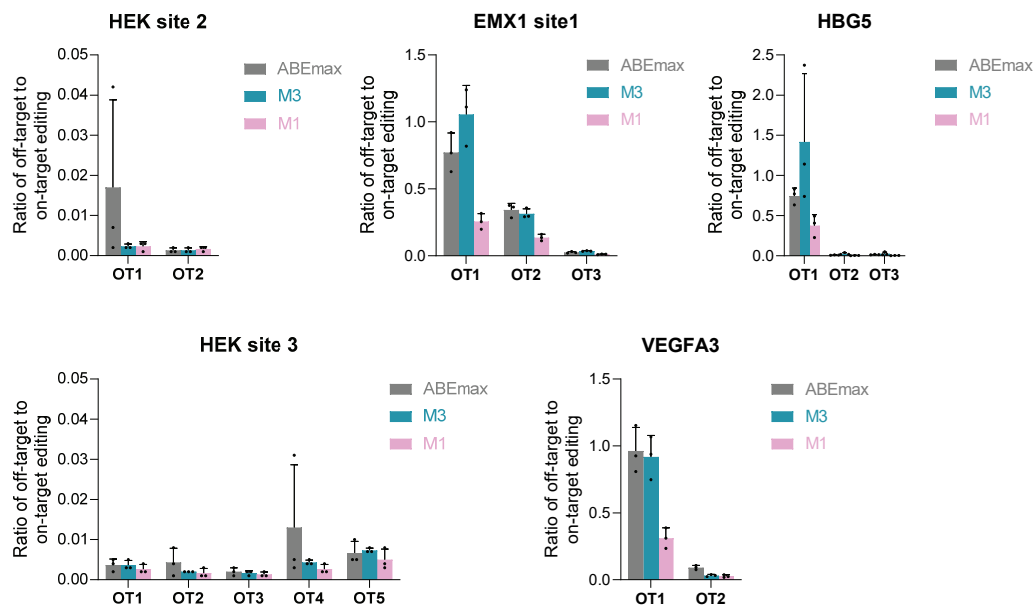
HEK3 OT4

HEK3 OT5

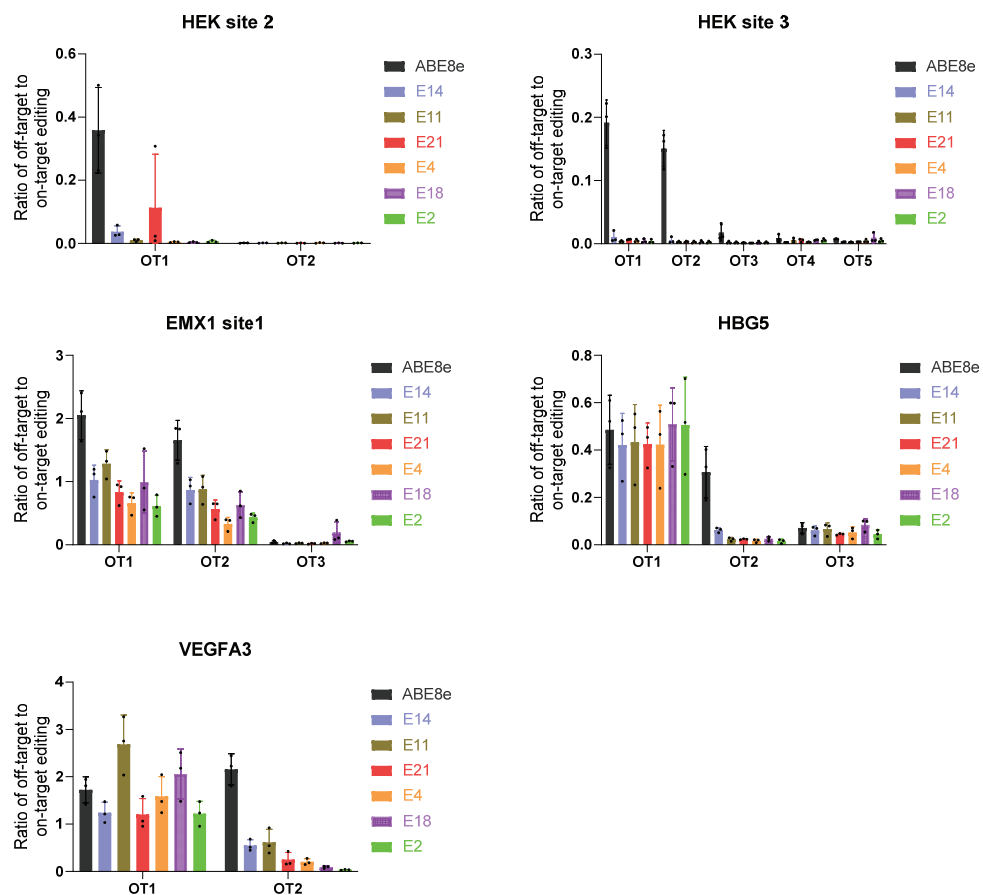
GGCCCAGACTGAGCACGTGATGG

CACCCAGACTGAGCACGT**G**CTGG**G**ACACAGAC**C**GGGCACGTG**A**GGG**A**GCT**C**AGACTGAGCA**A**GTG**A**GGG**A**GACCAGACTGAGCA**A**G**A**GAGGG**G**AGCCAGA**A**TGAGCACGTG**A**GGG

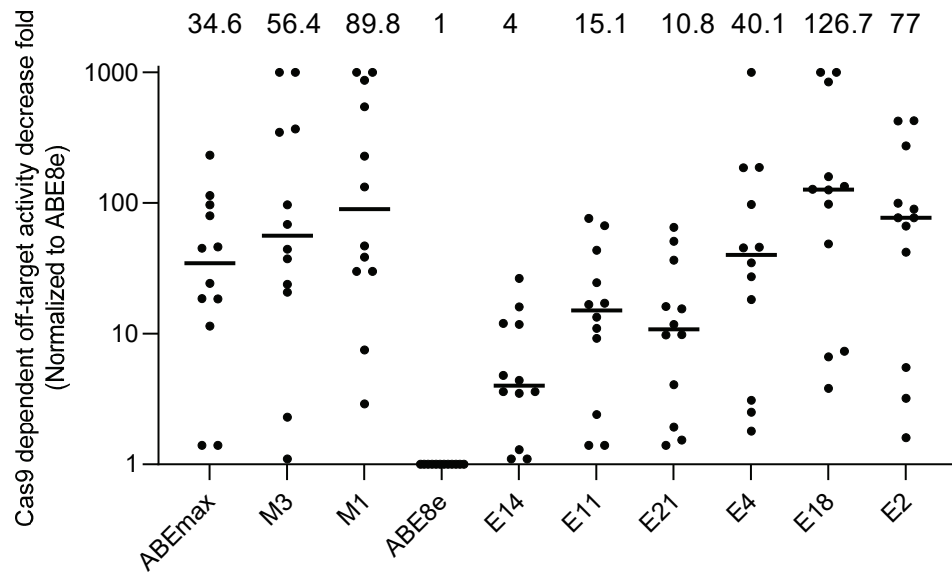
c



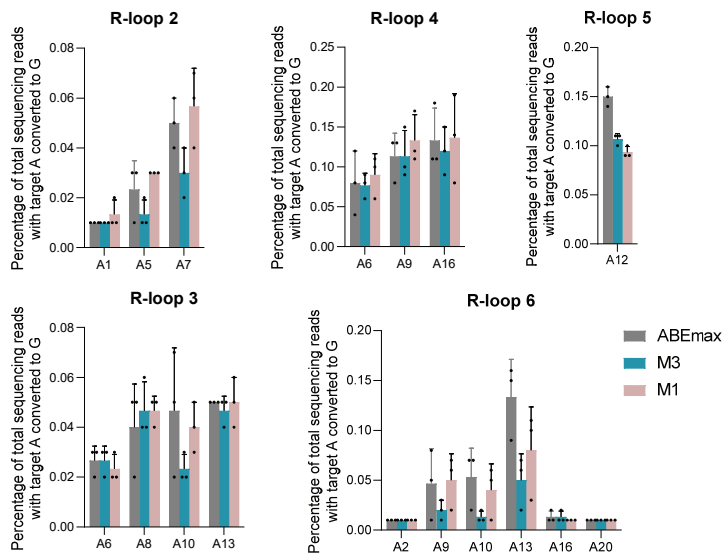
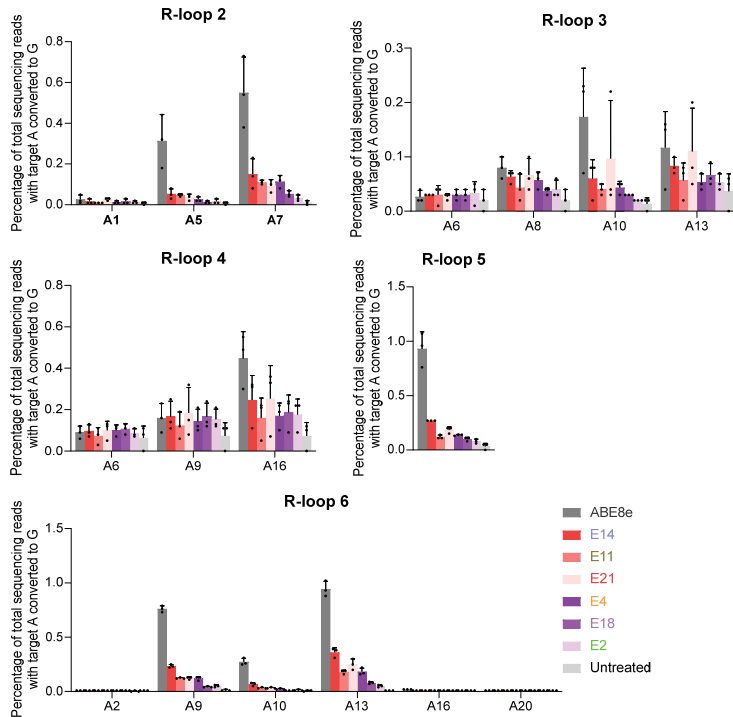
d



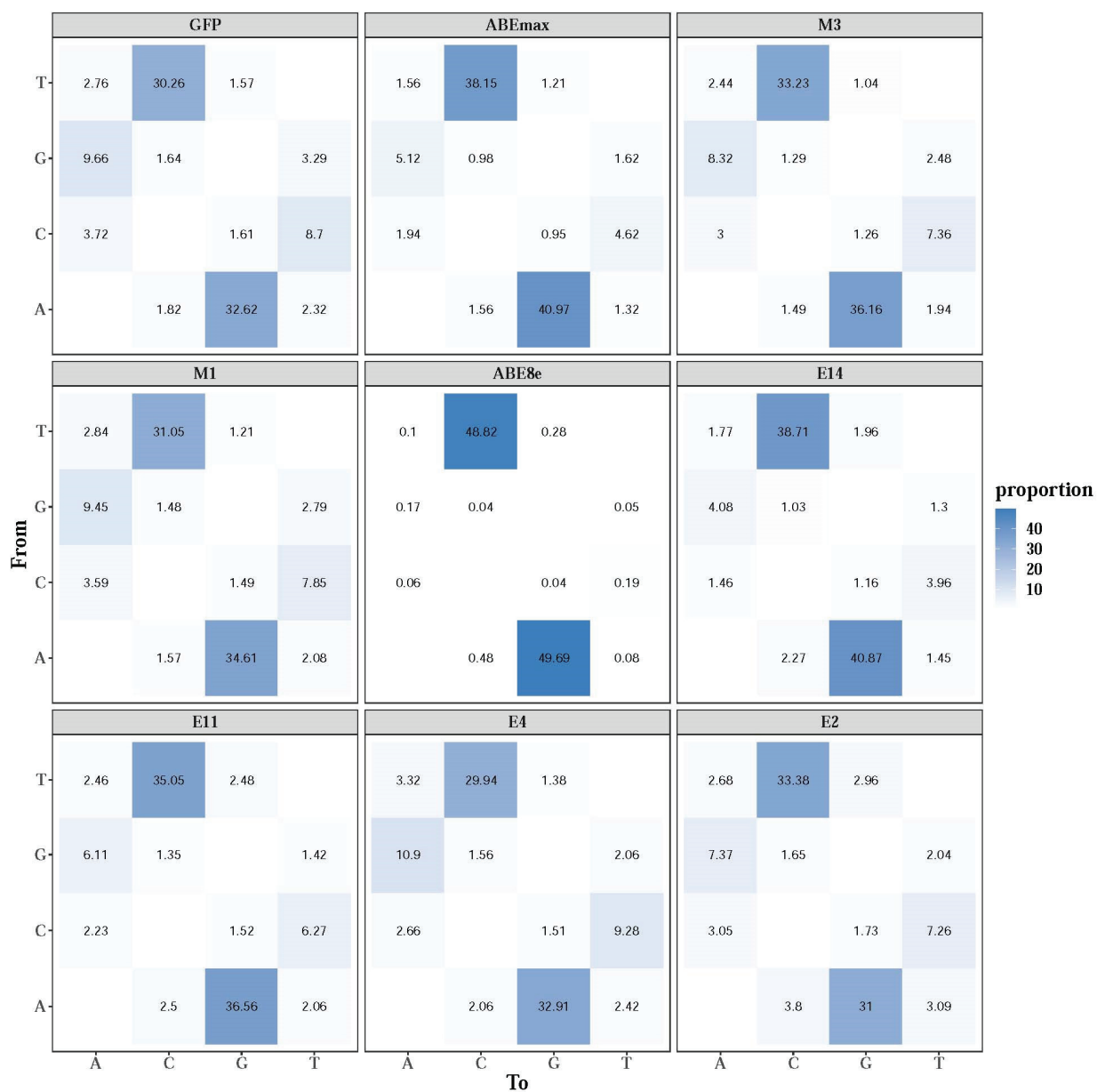
e



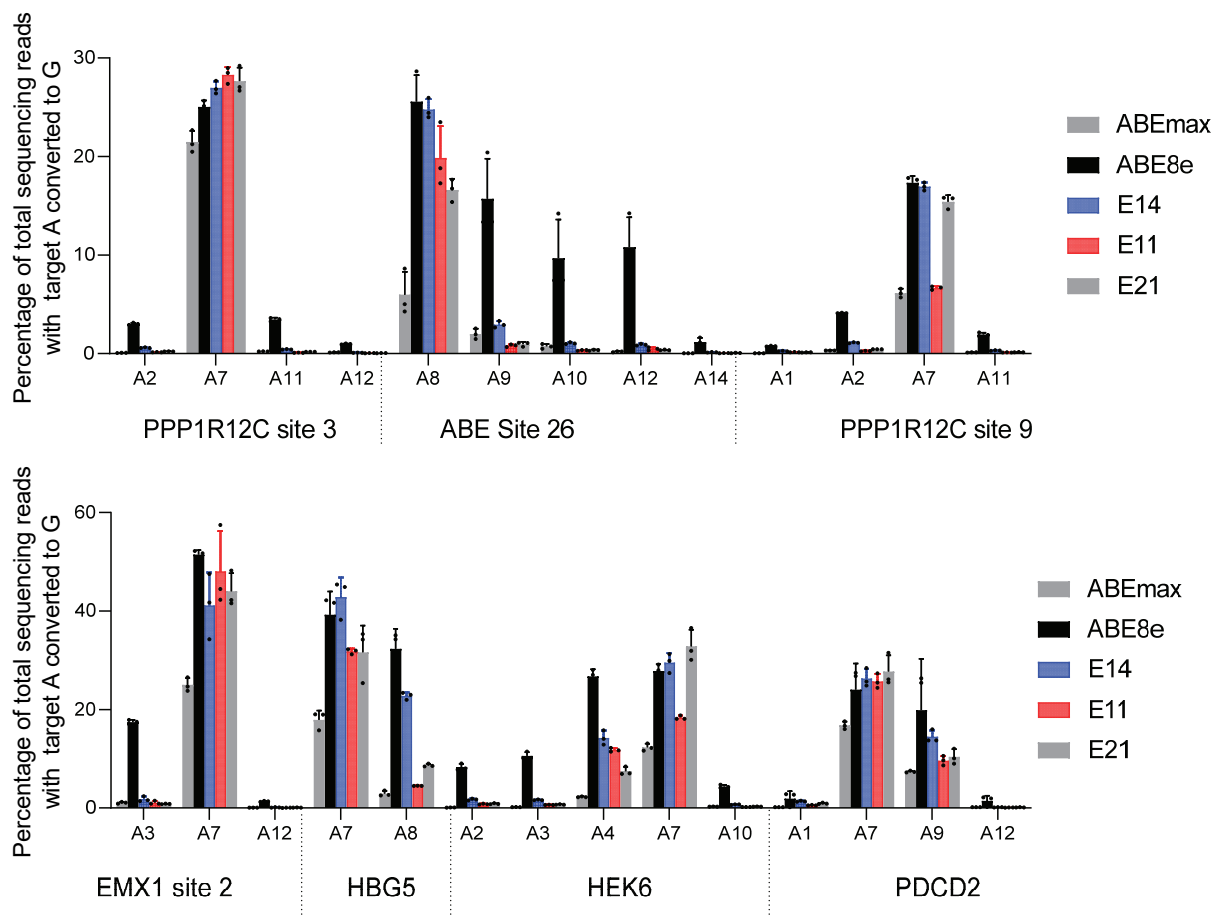
Supplementary Fig. 12| DNA off-target analysis of ABEmax, ABE8e and evolved variants. (a) Cumulative DNA off-target analysis of M3, M1 versus ABEmax at HBG site 5, VEGFA3, EMX1, and HEK293 site 2/3^{1,2}. (b) Cumulative DNA off-target analysis of E14, E11, E21, E4, E18, E2 versus ABE8e at HBG site 5, VEGFA3, EMX1, and HEK293 site 2/3. (c) The ratio of cumulative off-target to the highest on target editing of M3, M1 versus ABEmax based on a. (d) The ratio of cumulative off-target to the highest on target editing of E14, E11, E21, E4, E18, E2 versus ABE8e based on b. (e) Cas9-dependent off-target activity decrease fold number of evolved variants and ABEmax compared with ABE8e were calculated at all off-target sites but HEK site 2 OT2 and HEK site 3 OT4/5 (low value, no difference). The line represent median, and error bars represent the s.d. of at least three independent biological replicates (a, b). The sequence information of target sites and off-target sites were shown at right side (a, b).

a**b**

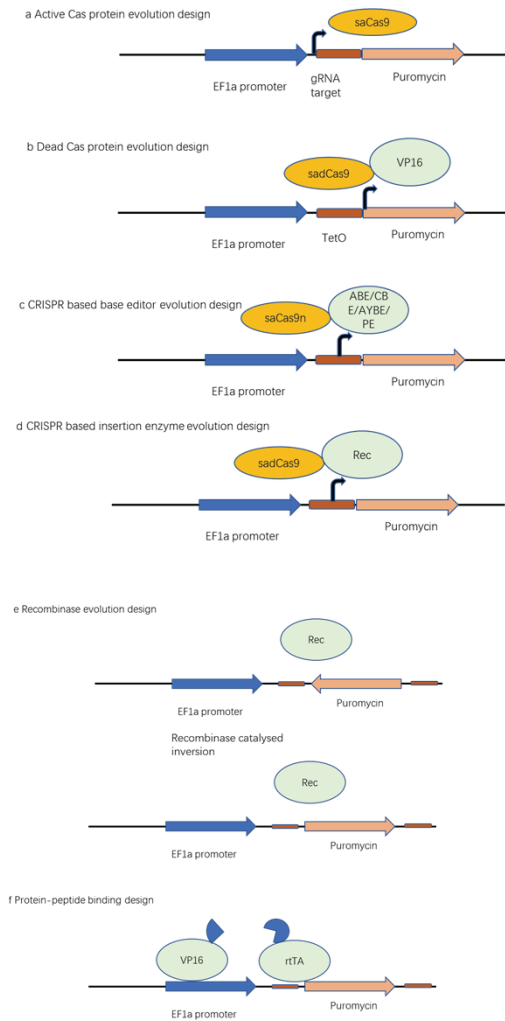
Supplementary Fig. 13| Cas9-independent deamination by adenosine deaminase in HEK293T cells. Cas9 independent off-target A•T-to-G•C editing frequencies of ABEmax and evolved variants (a), ABE8e and evolved variants (b) detected by the orthogonal R-loop assay at each R-loop site with dSaCas9 and a SaCas9 sgRNA. Each R-loop was performed by cotransfection of ABEmax, ABE8e, and evolved variants, an SpCas9 sgRNA targeting Site 3 with dSaCas9 and a SaCas9 sgRNA targeting R-loops 2–6, respectively.



Supplementary Fig. 14 | RNA off-target evaluation of evolved variants versus ABE_{8e} and ABE_{max}.
The number indicates the percentage of a certain type of mutation among all mutations.



Supplementary Fig. 15 | Base editing in HEK293T cells by E14, E11, E21 versus ABE8e and ABEmax. n=3 or 4 biologically independent experiments. Data were presented as mean±s.d. in histograms.



Supplementary Fig. 17 | Linkage of six kinds of protein activities to antibiotic expression. CDEM could be used to evolve several kinds of proteins including active Cas protein (a), dead Cas protein (b), base editor (c), enzyme for gene insertion (d), recombinase (e), binding protein(f).

Supplementary Sequences 1. Amino acid sequences for cytidine and adenosine deaminases of evolved base editor in this study.

N2-BE4

SSETGPVAVDPTLRRRIEPHEFEVFFDPRELKRETCCLYEINWGGRHSIWRHTSQNTNKHVEVNFIEKFTTERYFCPNT
RCSITWFLSWSPCGECSRAITEFLSRYPHVTLFIYIARLYHHADPRNRQGLRDLISSGVAIQIMTGQESGYCWRFVNY
SPSNEAHWPYPHPLWVRPVVLELYCIILGLPPCLNLRKQKQQLTFFTIALQSCHYQRLPPHILWATGLK

N7-BE4

SSETGPVAVDPTLRRRIEPHEFEVFFDPRELKRETCCLYEINWGGRHSIWRHTSQNTNKHVEVNFIEKFTTERYFCPNT
RCSITWFLSWSPCGECSRAITEFLSRYPHVTLFIYIARLHHHADPRNRQGLRDLISSGVTIQIMTEQESGYCWRFVNY
SPSNEAHWPYPHPLWVRPVVLELYCIILGLPPCLNLRKQKQQLTFFTIALQSCHYQRLPPHILWATGLK

N5-BE4

SSETGPVAVDPTLRRRIEPHEFGVFFDPRELKRETCCLYEINWGGRHSIWRHTSQNTNKHVEVNFIEKFTTERYFCPNT
RCSITWFLSWSPCGECSRAITEFLSRYPHVTLFIYIARLYHHADPRNRQGLRDLISSGVAIQIMTGQESGYCWRFVNY
SPSNEAHWPYPHPLWVRPVVLELYCIILGLPPCLNLRKQKQQLTFFTIALQSCHYQRLPPHILWATGLK

N12-BE4

SSETGPVAVDPTLRRRIEPHEFEVFFDPRELKRETCCLYEINWGGRHSIWRHTSQNTNKHVEVNFIEKFTTERYFCPNT
RCSITWFLSWSPCGECSRAITEFLSRYPHVTLFIYIARLYHHADPRNRQGLRDLISSGMTIQIMTEQESGYCWRFVNY
SPSNEAHWPYPHPLWVRPVVLELYCIILGLPPCLNLRGKQKQQLTFFTIALQSCHYQRLPPHILWATGLK

M3-ABEmax

SGVEFSHECWMRHALTLAKRAWDEREVPVGAVLVHNNRVIGEGWNRPIGRHDPTAHAEIMALRQGGLVMQNCR
LIDATLYVTLEPCVMCAGAMIHSRIGRVVFGARDAKTGAAGSLMDVLHHPGMNHRVEITEGILADECAALLSDFFR
MRRQEIKAKKKAQSSTDGSGSSGSSGSETPGTSESATPESSGGSSGSSSEVEFSHEYWMRHALTLAKRARDEREVP
VGAVLVHNNRVIGEGWNRRAIGLHDPTAHAEIMALRQGGLVMQNYRLIDATLYVTLEPCVMCAGAMIHSRIGRVV
GVRNAKTGAAGSLMDVLHYPGMNHRVEITEGILADECAALLCYFFRMPRQVFNAQKKAQSSTDGSGSSGSSGSE
TPGTSESATPE

M1-ABEmax

SEVEFSHEYWMRHALTLAKRAWDEREVPVGAVLVHNNRVIGEGWNRRAIGLHDPTAHAEIMALRQGGLVMQNYRL
IDATLYVTLEPCVMCAGAMIHSRIGRVVFGVRNAKTGAAGSLMDVLHYPGMNHRVEITEGILADECAALLCYFFRM
PRQVFNAQKKAQSSTDGSGSSGSSGSETPGTSESATPE

E2-ABE8e

SEVEFSHEYWMGHALTLAKRARDEREVPVGAVLVNNGGVIGEGWNRRAIGLHDPTAHAEIMALRQGGLVMQNYRLI
DATLYVTLEPCVMCAGAMIHSRIGRVVFGWNSKRGAAGSLMNALNHPGMNHRVEITEGILADECAALLCDFYR
MPRQVFNAQKKAQSSINSGGSSGSSGSETPGTSESATPE

E4-ABE8e

SEVEFSHEYWMRHALTLAKRARDEREVPVGAVLVNDRVIGEGWNRRAIGLHDPTAHAEIMALRQGGLVMQNYRLI
DATLYVTLEPCVMCAGAMIHSRIGRVVFGWRNSKRGAAGSLMNVNYPGMNHRVEITEGILADECAALLCDFYRM
PRQVFNAQKKAQSSINSGGSSGSSGSETPGTSESATPE

E11-ABE8e

SEVEFSHEYWMRHALTLAKRARDEREVPVGAVLVNDRVIGEGWNRRAIGLHDPTAHAEIMALRQGGLVMQNYRLI
DATLYVTLEPCVMCAGAMIHSRIGRVVFGWRNSKRGAAGSLMNVNYPGMNRRVEITEGILADECAALLCDFYRM
PRQVFNAQKKAQSSINSGGSSGSSGSETPGTSESATPE

E14-ABE8e

SEVEFSHEYWMRHALTLAKRARDEREVPVGAVLVNNGGVIGEGWNRRAIGLHDPTAHAEIMALRQGGLVMQNYRLI
DATLYVTLEPCVMCAGAMIHSRICRVVFGWNSKRGAAGSLMNVNYPGMNHRVEITEGILADECAALLCDFYRM
PRQVFNAQKKAQSSINSGGSSGSSGSETPGTSESATPE

E18-ABE8e

SEVEFSHEYWMRHALTLAKRARDEREVPVGAVLVLNNRVIGEGWNRAIGLHDPTAHAEIMALRQGGPVMQNYRL
IDATLYVTFEPCVMCAGAMIHSRIGRVVFGWRNSKRGAAAGSLMNVPNYPGMNHRVEITEGILADECAALLCDFYR
MPRQVFDAQKKAQSSINSGGSSGGSSGSETPGTSESATPE

E21-ABE8e

SEVEFSHEYWMRHALTLAKGARDEREVPVGAVLVLNNRVIGEGWNRAIGLHDPTAHAEIMALRQGGLVMQNYRL
IDATLYVTFEPCVMCAGAMIHSRIGRVVFGWGNNSKRGAAAGSLMNVLNYPGMNHRVEITEGILADECAALLCDFYR
MPRQVFNAQKKAQSSINSGGSSGGSSGSETPGTSESATPE

Supplementary Table 1. Materials used

REAGENT or RESOUCCE	SOURCE	IDENTIFIER
Lipofectamine 2000	Thermo fisher	11668019
Lipofectamine 3000	Thermo fisher	L3000-015
EZtrans	life iLAB, China	AC04L092
293T cells	National Collection of Authenticated Cell Cultures	GNHu44
Puromycin	Solarbio	58-58-2
Hygromycin B	APExBIO	A2515
Blasticidin	Solarbio	3513-03-9
Zeocin	Solarbio	11006-33-0
DMEM	gibco	C11995500BT
Opti-MEM	gibco	31985-070
Penicillin/Streptomycin	Thermo fisher	15140163
FBS serum	ExCell Bio	FSP500
Non-essential amino acids	gibco	11140-050
0.45uM filters	Millex	SLHVR33RB
6/12/24 well Pates	Biofil	TCP-010
Flask	CORNING	430641
Trypsin	gibco	25200-072
Stbl3 Chemically Competent E. coli	Homemade	
Lenti-X Concentrator	Takara	631231
Plasmid extraction	Magen	P1001-03/P1003-03
GeneJet Genomic DNA Purification kit	Thermo fisher	K0721
Phanta Max super-fidelity DNA polymerase	Vazyme	P505

Variants	Editing activity	Editing window length	PAM	TadA state	The base of variant generation	Mutations	Ref
ABE8.8-m/d	>ABEmax	>ABEmax	NG/NGG	Monomer	ABEmax	ABE8.8: Y123H, Y147R,Q154R	⁶
ABE8.13-m/d	>ABEmax	>ABEmax	NG/NGG	Dimer	ABEmax	ABE8.13: I76Y, Y123H, Y147R,Q154R	⁶
ABE8.17-m/d	>ABEmax	>ABEmax	NG/NGG	Monomer	ABEmax	ABE8.17: V82S,Q154R	⁶
ABE8.20-m/d	>ABEmax	>ABEmax	NG/NGG	Dimer	ABEmax	ABE8.20: I76Y, V82S, Y123H, Y147R,Q154R	⁶
NG-ABE9e	≈ NG-ABE8e	<NG-ABE8e	NG	Monomer	ABE8e	R111T	⁷
ABE8e-N108Q	≈ ABE8e	<ABE8e	NGG	Monomer	ABE8e	N108Q	⁸
ABE9	<ABE8e	<ABE8e-N108Q	NGG	Monomer	ABE8e	N108Q, L145T	⁸
ABE8r	≈ ABE8e	>ABE8e	NGG	Monomer	ABE8e	R47K,I76Y,V82S,N108G,K110R,R111H,A114V ₉ ,M126I,N127K,D147R,Q154R	
E4-ABE8	≈ ABEmax	<ABEmax	NGG	Monomer	ABE8e	N38D,L121P	This study
E2-ABE8	≈ ABEmax	<ABEmax	NGG	Monomer	ABE8e	R13G,N38G,R39G,R107G,V120A,Y123H	This study
E11-ABE8	≈ ABE8e	ABEmax<E11-ABE8<ABE8e	NGG	Monomer	ABE8e	L121P,H128R	This study
E14-ABE8	≈ ABE8e	ABEmax<E14-ABE8<ABE8e	NGG	Monomer	ABE8e	N38G,R39G,G100C,R107G,S172G	This study

Supplementary Table 2. | Comparison of ABE variants. TadA Variants were compared with either that of ABEmax or ABE8e.

	CDEM	PACE	Directional screening
Evolution environment	Eukaryotic	Prokaryotic	Eukaryotic
Continuou evolution or non-continuou	Continuou	Continuou	Non-continuou
Length restriction	No	4.4 kb	No
Biosafety level	No	Biosaftey level 2 lab	No
Adjustment of experimental parameters	Simple	Complicated	Simple
Evolution time	About 2 months	About 1 week	Dependent on screening times
Reference	This study	³	⁴

Supplementary Table 3. | Comparison our CDEM to PACE and directional screening in human cells.

Supplementary protocol: Evolution of BE3 as an example

Section A-Plasmid preparation

Designed gRNAs distribute evenly on gene to be evolved. For the possible mutational tolerance caused by neutral evolution, designing more gRNAs is an efficient approach.

For evolving protein with long size, it will help through restricting mutations to regions of the gene that are expected to show improvement according to bioinformatics analysis and structure information.

Base editing system to be evolved:

pLenti-FNLS-P2A-Puro (expressed stably)

Lv-gBlast-Hygro (expressed either stably or transiently)

Mutagenesis system:

pTE-dCas9-NeoR (expressed transiently)

pGL-MS2gRNA-MS2-AID-Δ-BeoR (expressed transiently)

Selection plasmid

Lv-GTG-blast (expressed transiently)

Section B- Stable Cell line preparation

293T stably expressing BE3 and gRNA targeting deficient Blastocidin were made by transducing 293T cells with lentivirus (pLenti-FNLS-P2A-Puro) followed by Puromycin (1μg/ml) (Solarbio) selection and lentivirus (Lv-gBlast-Hygro) followed by Hygromycin B (400μg/ml) (APExBIO) selection.

Lentiviral Vector (LV) production (Any other protocol can produce lentivirus also OK. For example: <http://www.addgene.org/protocols/lentivirus-production/>)

Day 1: seed 293T cells (in DMEM + FBS + non-essential amino acids) at 70-90% confluence.

Day 2: transfection

1. Prepare Mix A: Mix lentivector, pAX2 and VSV-g at the ratio:1.64:1.3:0.72 in Opti-MEM.

2. Prepare Mix B: Mix Lipofectamine2000 and Opti-MEM according to the instruction.

3. Combine Mix A + B, incubate at room temperature for 5-20 minutes.

4. Carefully add the A+B mixture to the cells.

Day 3: Replace the culture medium with fresh Opti-MEM (carefully, to avoid losing the cells; If you want, you can add pen/strep to the Opti-MEM)

Day 4: Collect the Lentivirus-containing culture medium. Remove cells by low-speed centrifugation (1500 rpm for 5 min). Transfer the supernatant to a new tube and filter with 0.45 μm filter. Store aliquots at -80°C.

Lentivirus Transduction and positive cells selection

#To get single copy number cells, calculating the transduction titer is one approach, it is also possible to transduce 293T cells with different titer of lentivirus.

#Lenti-X Concentrator can be used to concentrate the lentivirus if large transgene was used.

Day 1 Plate 293T cells and incubate at 37°C, 5% CO2 overnight.

Day 2 Cells should be approximately 70% confluent when transduction. Thaw aliquots of lentiviral stocks on ice. Add proper volume of the lentiviral stocks into 293T cells at desired MOI (Multiplicity of Infection).

Day 5/6 Passage the cells, and add the corresponding antibiotic. The cells should be cultured in corresponding antibiotic for 7 days (Puromycin and Blasticidin), or 10 day (Hygromycin B)

Section C-CDEM

#After five rounds of mutagenesis and selection, the cells were collected and saved for genomic DNA extraction.

#The maximal antibiotic concentration can be around 200-300µg/ml.

Day 0 Plate 293T cells and incubate at 37°C, 5% CO₂ overnight. Cells should be approximately 70% confluent when transfection.

Day 1 Transient transfection was performed on 293T cells with mutagenesis plasmids (pTE-dCas9-NeoR and 10 gRNAs (backbone on pGL-MS2gRNA-MS2-AID-Δ-BeoR) targeting the cytidine deaminase)

Day 5 Four days after transfection, the cells were spread.

Day 6 Transient transfection was performed with selection plasmids (Lv-GTG-blast).

Day 8 Two days later, the cells were passaged and Blasticidin was added at same time with increasing concentration (start from 10µg/ml).

Day N When the cells grow full, star new round of evolution.

Section D-Clone Out Your Evolved Transgene

Genomic DNA extraction:

The cells were harvested after antibiotic selection and the genomic DNA extraction was performed according to manufacturer's instructions (Thermo Scientific).

Amplify and Subclone the Transgene

1 Amplify the evolved gene.

Follow all manufacturer's instructions and protocols. Standard amplification.

2 Gel purify the resulting product for the correct sized band.

3 Digest pLenti-FNLS-P2A-Puro and the PCR products with BamH1 and Apa1.

4 Ligate the backbone and PCR products together with T4 DNA Ligase

5 Transform ligated plasmid to Stbl3 Chemically Competent E. Coli

6 Plate cells on selective media

7 Isolate, culture, and DNA prep as needed

If many rounds of selection have been performed only a few colonies will be necessary.

8 Sequence the evolved gene region and analyze for consensus.

9 Perform an appropriate functional assay to test the resulting mutants.

Supplementary Discussion

Our evolved E14/E11/E21-ABE8e variants were used to correct the mutation on CFTR gene that is more efficient and precise than ABE8e and ABEmax (Fig. 3i). Similar performance was observed on the other 6 sites (Fig. 3h). Variant E14/11/E21 showed lower off-target activity compared with ABE8e according to previous results (Fig. 3e-g). Collectively, E14/E11/E21-ABE8e variants have great potential for therapeutic application in which high precision editing and low off-target activity is required. The other variants with their specific characters can also provide more options for specific genetic disease therapies. For example, our evolved N7-BE4 has narrower editing window, higher editing precision and lower off-target activity compared with previous reported YE1-BE4, and E11-ABE8-ABE8e typified by shifted editing window, as well as, M3-ABEmax, M1-ABE8, E4-ABE8 and E2-ABE8 with comparable or slightly lower editing activity but narrower editing window at varying degree in comparison with ABEmax. These variants evolved by CDEM substantially advances the capabilities of ABEs and CBEs by improving the targeting spectrum and precision, as well as high specificity and product purity for base editing applications.

Through our CDEM, we acquired a M1-ABE variant with TadA as monomer evolved from ABEmax with TadA as heterodimer, which is different from TadA of ABE8e evolved from monomer TadA-7.10⁵. As Fig 1g show, TadA monomer is a recombination of TadA heterodimer of ABEmax. These may indicate another advantage of our CDEM, deletion or recombination can be introduced through CDEM.

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- 4 Xie, H. *et al.* High-fidelity SaCas9 identified by directional screening in human cells. *PLoS biology* 18, e3000747, doi:10.1371/journal.pbio.3000747 (2020).
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